European Network of
Transmission System Operators
For Electricity

Functional Specification

Energy Communication Platform
Version 4.16.0

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Energy Communication Platform, version 4.16.0

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2. Executive Summary

This document contains the description of Energy Communication Platform from the user point of view. It specifies in detail the behaviour of the ECP software. The document assumes familiarity with the ECP basic goals and high-level concepts. The document is targeted to the MADES Working Group and to vendor's developers.

More implementation-specific and detailed information can be found in the ECP System Design document.

This document is targeted at Business Users, Administrators and Developers.

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3. Revision History

Version	Date	Author	Description
4.0.0	24.3.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 1 version
4.0.1	13.6.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 2 Iteration 1 version: Introduce the Component Directory
4.0.2	10.7.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 2 Iteration 2 version: Describe the message paths conflicts and validity
4.0.3	22.7.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 2 Iteration 3 version: Describe the use cases related to Component Directory
4.0.4	12.8.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 2 Iteration 4 version: Revision comments processed, ECP logo added
4.0.5	5.9.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 2 Iteration 5 version: Revision comments processed
4.1.0	13.4.2017	Jiří Neuman	ECP 4.2 version: Added support for multiple networks
4.1.4	05.02.2018	Laura Betcherová	Documentation review
4.2.0	21.03.2018	Josef Brož	Version update
4.3.0	25.06.2018	Josef Brož	CD - extended monitoring
4.4.0	14.06.2019	Josef Brož	Version update
4.5.0	22.08.2019	Jiří Kakrda	Version update
4.6.0	05.11.2019	Jiří Kakrda	Version update
4.6.1	18.03.2020	Anna Langrová	Version update
4.7.0	28.06.2020	Lukáš Voříšek	Change time zone, Endpoint reinitialize, CD synchronization update
4.7.1.	31.07.2020	Anna Langrová	Version update

4.7.2	16.11.2020	Jiří Kakrda	Component Directory Synchronizer
4.8.0	10.03.2021	Tomáš Vondra	UC Components update
4.8.1	09.07.2021	Lukáš Voříšek	Version update
4.8.2	03.12.2021	Daniel Janota, Jiří Kakrda	Project and Environment names, GUI screens update, formatting
4.9.0	26.05.2022	Jiří Kakrda	Background Jobs, Inbox and Outbox – organization, status description
4.10.0	07.11.2022	Daniel Janota, Jiří Kakrda	Hibernate, components remove button, UC Settings – Names section, UC Settings – runtime configuration editing, components export button, Dashboard – Readiness tile
4.11.0	16.05.2023	Jiří Kakrda	Fixes of references to Administration Guide, correct description of header colours
4.12.0	27.10.2023	Jiří Kakrda	Version update
4.14.0	10.06.2024	Jiří Kakrda	Artemis
4.15.0	21.10.2024	Jiří Kakrda	Database Backup GUI
4.16.0	06.05.2025	Jiří Kakrda	Version update

4. Functional Design

4.1. Features Overview

The purpose of Energy Communication Platform is to provide message delivery capabilities with following additional key features:

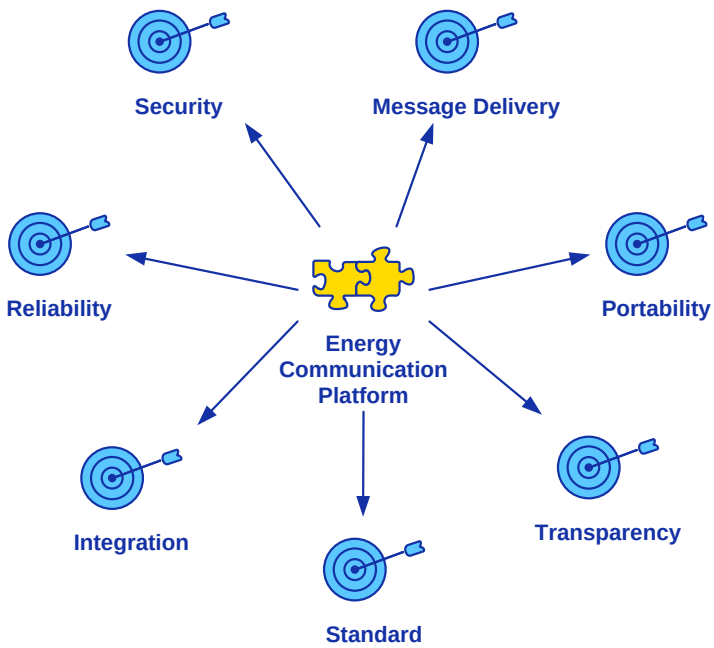

Figure 1: Key Features

Security – Only recipient of the message can read the message content. The sender of any message can be unambiguously authenticated.

Reliability – The message delivery is guaranteed.

Integration – The ECP functionality for sending and receiving messages can be integrated with wide variety of technologies.

Standard – ECP is implemented based on MADES Technical Specification of IEC (TR 62325-503) that is in the process to be published as an International Standard in 2018. MADES which defines the technical aspects of communication between entities in the energy sector.

Transparency – Any message transported by ECP can be tracked down to gather trustworthy information about the state of delivery

Portability – The ECP can be installed on most of widely-used operation systems

Energy Communication Platform is internally composed of four main components – Endpoint, Internal Broker, Component Directory and Broker.

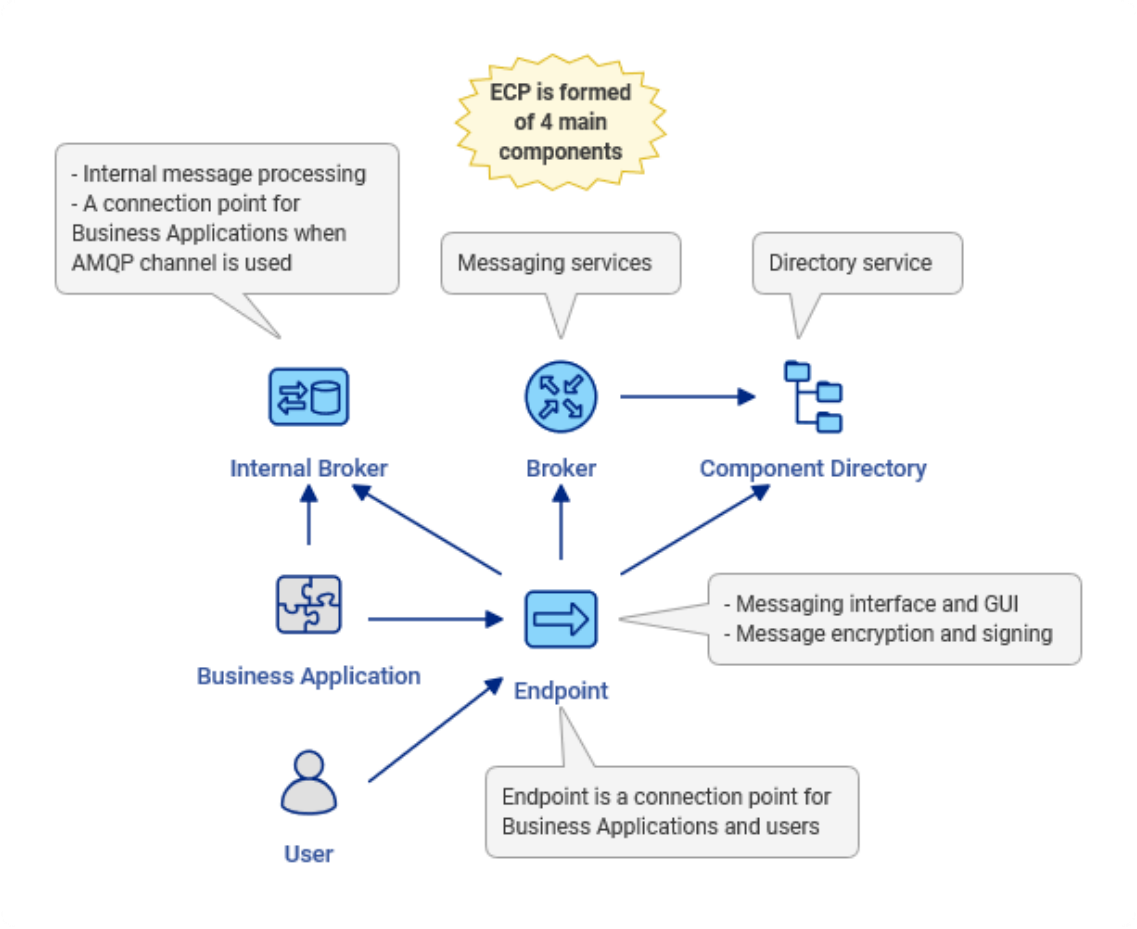

Figure 2: Main Components

- From the users' point of view, the crucial component is the Endpoint, which provides the user interface for sending and receiving messages and the API for integration with business applications.
- The Internal Broker provides messaging services for the internal message processing in Endpoint. Endpoint must be always connected to Internal Broker to be able to process messages. Multiple Endpoints can be connected to one Internal Broker instance. Business Application can also send and receive messages from Internal Broker in case the AMQP integration channel is used.
- The "Component Directory" component serves as a central part of ECP network managing the information about all existing ECP component – Endpoints, Brokers and other Component Directories.
- The Broker component provides the messaging services to the Endpoints.

The detailed description of the functions provided by ECP and the role of each ECP component are described in the next sections.

4.2. Messaging

Messaging is the key functionality of ECP. The messaging process begins with a business application (BA) uploading a message for transport through the ECP network and ends with the reception of the message by a target business application. Several components within ECP network are involved in the process of delivering the message to its destination.

The complete path of a message through ECP network is shown on the diagram below and the description of each step is provided.

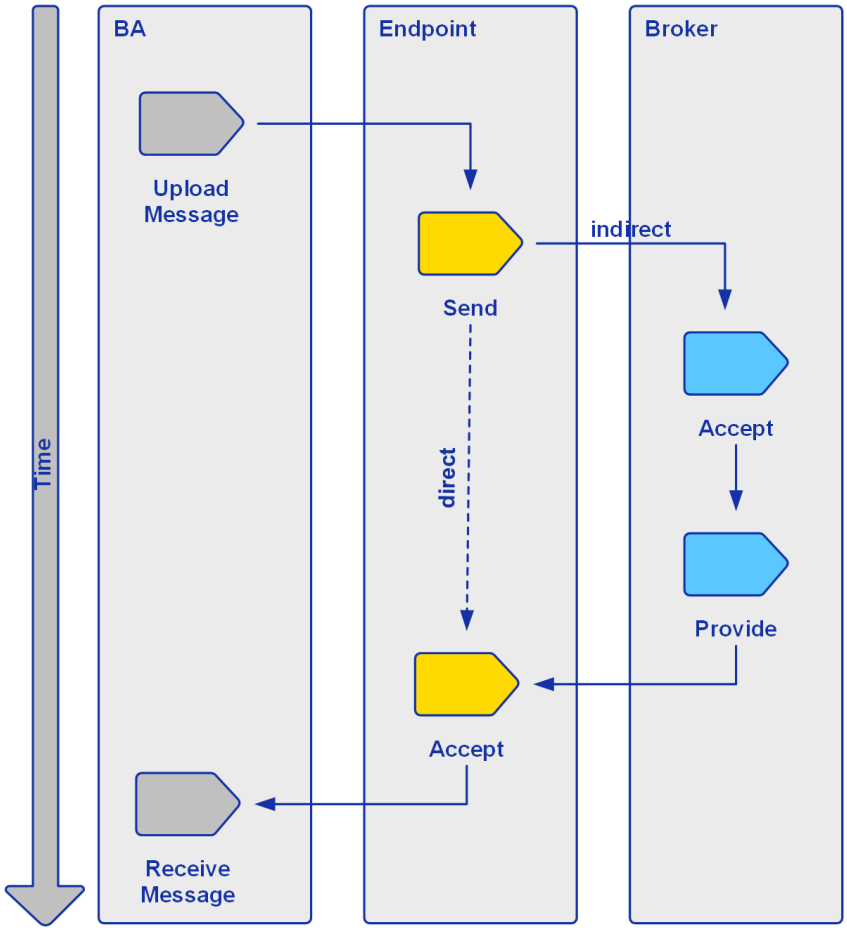

Figure 3: Messaging process overview

1. A business application uploads a message to the sender Endpoint (or to Internal Broker connected to ECP Endpoint in case of AMQP channel).
2. The sender Endpoint sends the message to the recipient Endpoint Internal Broker (direct transport) or to a Broker (indirect transport).
3. The Broker accepts the message from the sender Endpoint (only indirect transport).
4. The Broker provides the message to recipient Endpoint (only indirect transport).
5. The recipient Endpoint accepts the message.

6. The target Business application receives the message from the recipient Endpoint.

4.2.1. Endpoint - Send Message

The messaging process in ECP starts with an ECP Endpoint accepting a message from a user or a business application (BA) and sending it towards the receiver through a predefined message path. This step is the same for both the direct and the indirect transport pattern.

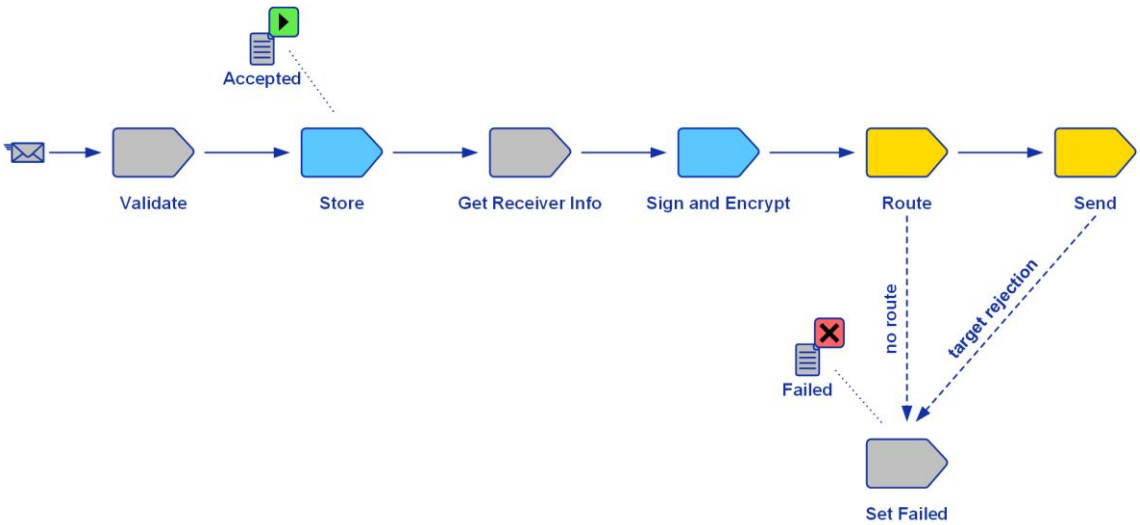

Figure 4: Messaging - Endpoint sends a message

1. The sender Endpoint validates that the message has a correct format and contains all required information (e.g. receiver, message type).
2. The sender Endpoint stores the message.
3. The sender Endpoint gets security information (e.g. encryption certificate) about the recipient Endpoint of the message. It gets such information from local Component Directory data.
4. The sender Endpoint signs the message content and relevant metadata and optionally compresses the content and encrypts the message content.
   > The acknowledgments are signed but they are not compressed nor encrypted.
5. The sender Endpoint selects the message route through ECP network according to the configured message paths.
   > If there is no valid message path according to the message type, the Endpoint moves the message to the failed state.

> If the selected path is a direct path to the recipient Endpoint and if the sender Endpoint does not allow incoming direct transport, the system moves the message to the failed state. The reason is that the recipient Endpoint will not be able to send back the acknowledgement using the reverse route, which is then also direct transport.

6. The sender Endpoint sends the message to the next ECP component through the selected route, i.e. the receiver Internal Broker when transport is direct, otherwise an intermediate Broker.

> If the target ECP component rejects the message due to security or validation error, the sender Endpoint moves the message to the failed state.

Note: the term "*message type*", which is used all along the document, exactly refers to the term "*business-type*" as defined in MADES.

4.2.2. Broker - Accept Message

In case of indirect transport, an intermediate Broker executes the second step of the messaging process. The Broker accepts the message from the sender Endpoint and stores it.

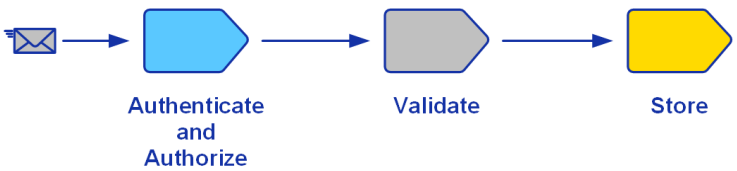

Figure 5: Messaging - Broker accepts a message

1. The Broker act as a server: it authenticates the connecting sender Endpoint and authorizes the upload of the message.
2. The Broker validates the message and checks it against transport rules defined for the Broker (further described in the document).
3. The Broker stores the message so that the recipient Endpoint can pick it up.

The Broker reports any security related errors or failed validations directly to the client (e.g. the sender Endpoint).

4.2.3. Broker - Provide Message

In case of indirect transport, the Broker also executes the third step of the messaging process. The Broker provides the message to the recipient Endpoint.

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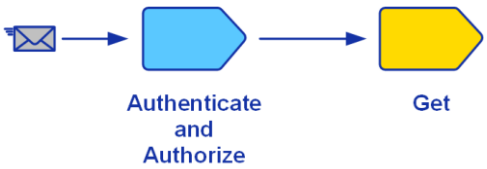

Figure 6: Messaging - Broker provides a message

1. The Broker acts as a server: it authenticates the connecting recipient Endpoint and authorizes the access to its messages.
2. The Broker fetches the message and provides it to the Endpoint.

The Broker reports any security related errors directly to the client (e.g. the recipient Endpoint).

4.2.4. Endpoint - Accept Message

The last step in messaging process applies for both the direct and indirect transport patterns. The recipient Endpoint accepts the message either directly from a sender Endpoint or indirectly from an intermediate Broker.

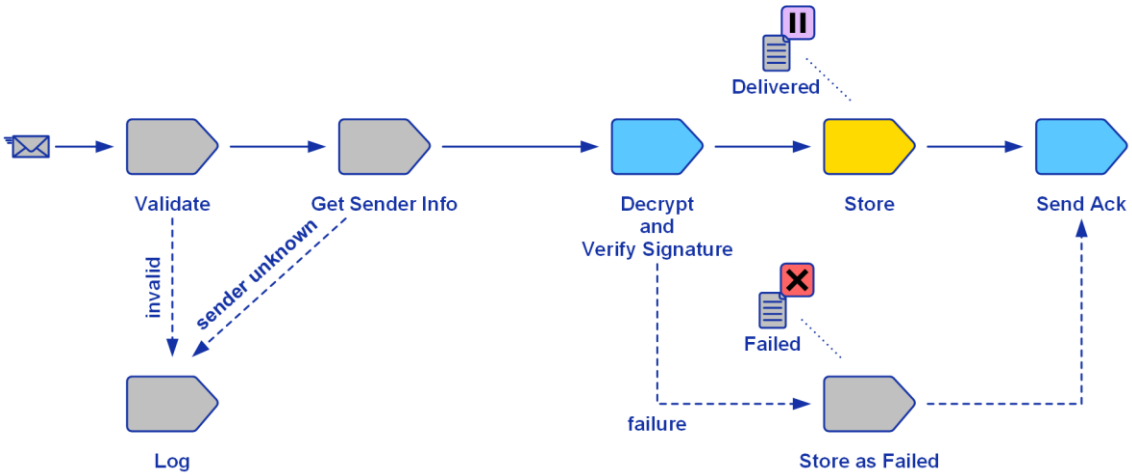

Figure 7: Messaging - Endpoint accepts a message

1. The recipient Endpoint accepts a message.
2. The recipient Endpoint validates the format of the message.
   - If the message has invalid format, the Endpoint logs the error and discards the message.
3. The recipient Endpoint gets security information about the sender of the message (e.g. signing certificate). It gets such information from local Component Directory data.

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4. The recipient Endpoint decrypts and decompresses the message content and verifies the signature of the sender Endpoint.
   > If the decryption or the verification of the signature fail, the Endpoint stores the message and marks it as failed.
5. The recipient Endpoint stores the message and marks it as delivered.
6. When the received message is a business message, which means it is not an acknowledgement, the recipient Endpoint sends an (technical) acknowledgment back to the sender Endpoint.

Note – The "technical acknowledgement" notifies that the recipient Endpoint or a recipient's business application (BA) has accepted the business message. It shall not mean more that the message has been technically and securely delivered to the recipient Endpoint or a BA of the recipient party. So far, the content of the message has not been analysed. The probable and further analysis may result in a "functional acknowledgment", the document being then accepted or rejected according to the business rules. Such a functional acknowledgement can even be a new message that the ECP network will deliver as a new business-message to the sender of the original message.

4.2.5. Message Lifecycle

Every message sent through the ECP network has a well-defined lifecycle, described below. There are two types of lifecycles – full lifecycle for the business messages and a short lifecycle for the technical acknowledgments.

The full lifecycle consists of several possible states that the message can be in, some of them being alternative.

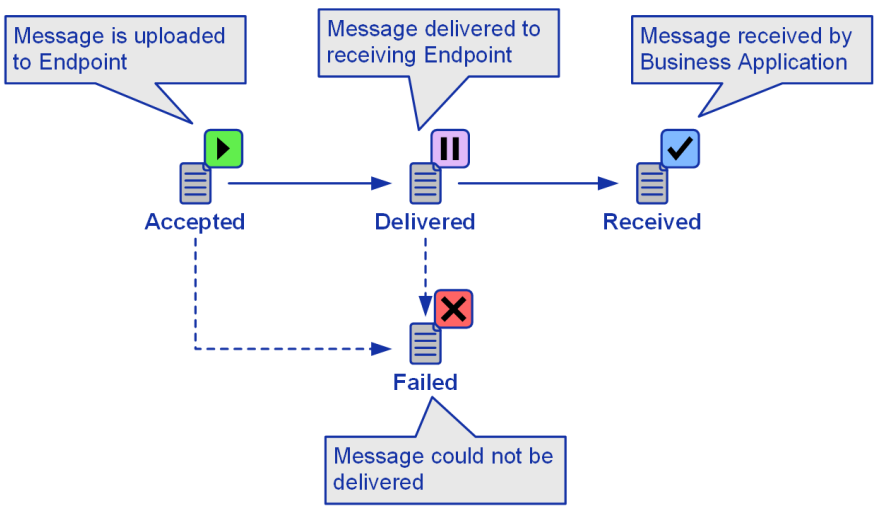


Figure 8: Message lifecycle

It is important to know that the state of a message is not an attribute of the message transported with it. It is information that each component that sees the message associates to it. As a message is passing through several components, each component associates a state to the message. Not all these states may have the same value at the same time, but ECP ensures that they are all eventually consistent.

The presented lifecycle of a message is as viewed and managed by one component.

The message moves in the lifecycle based on conditions.

1. Accepted (Initial) – The initial state of a message is Accepted. Any message uploaded to ECP by a business application is in the Accepted state.
2. Delivered – When the message is delivered to the recipient Endpoint, it is moved to the "Delivered" state. In this state, the message waits for reception by a business application.
3. Received (Final) – The final state of the message is the Received state, signalling that the message was received by a business application.
4. Error (Alternative final) – The message reaches the error state when it cannot be delivered whatever reason. Typically, this is due to an unknown receiver, security errors (e.g. invalid signature) or message expiration. This state is a final state.

The lifecycle of an acknowledgment is a shortened version of the full lifecycle.

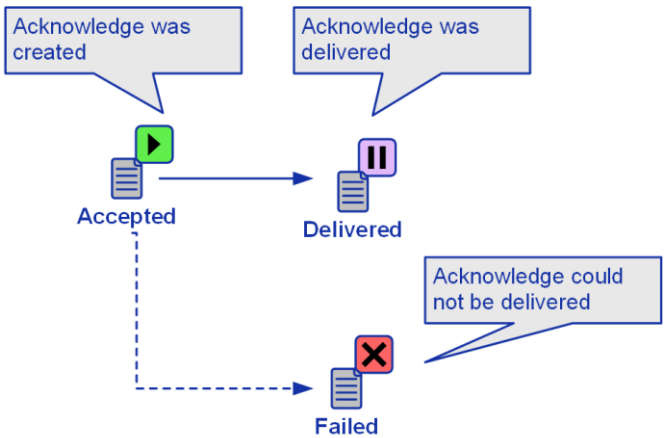


Figure 9: Acknowledgment lifecycle

The meaning of the states is the same as for a business message, but the Received state is not used.

4.3. UC Check Message Delivery Status

The Check Message Delivery Status use case is used to get information about the delivery status of a given message. The sender Endpoint receives a check status request from a Business Application; it elaborates a delivery status from the received acknowledgments related to the requested message and returns the status to the requesting Business Application.

4.3.1. Activity Diagram

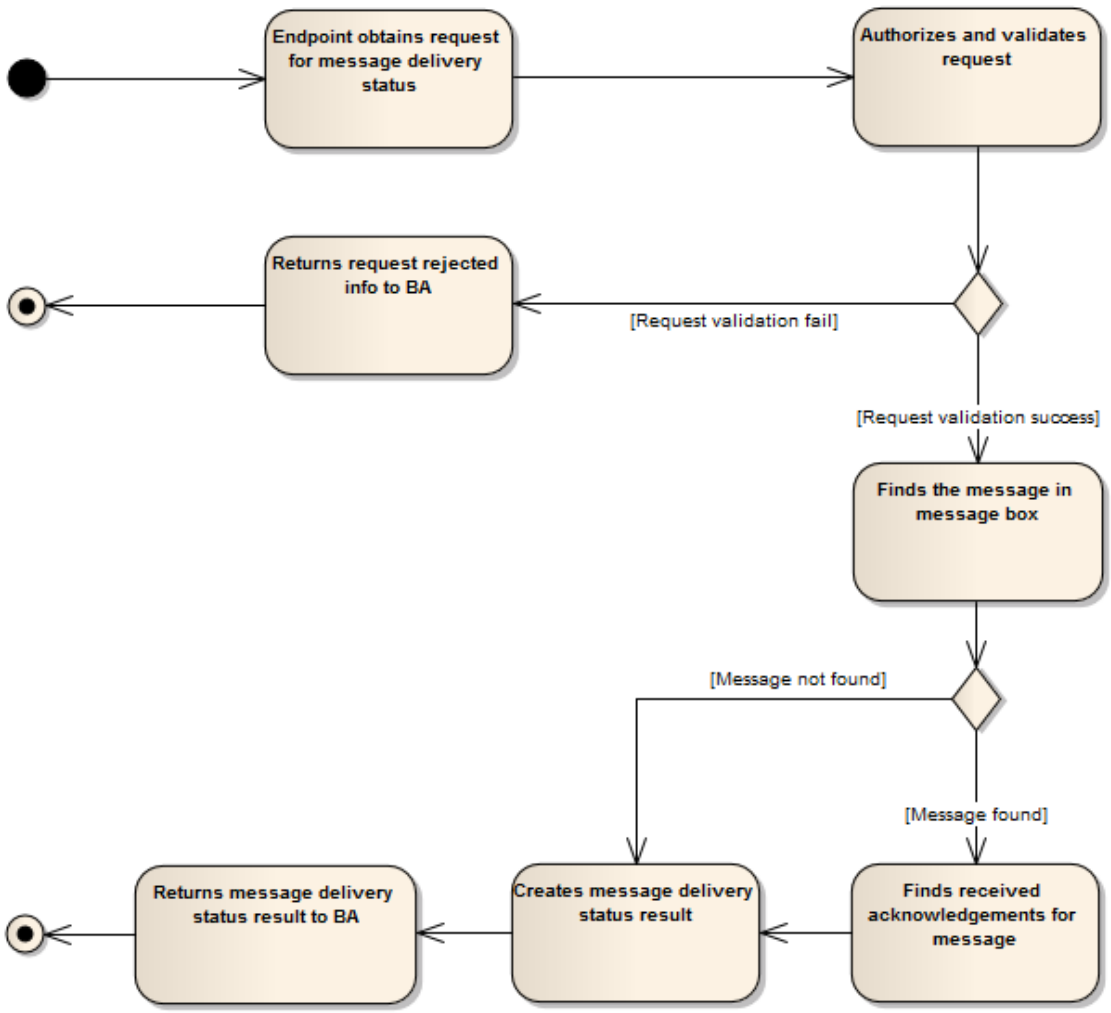


Figure 10: Check Message Delivery Status

4.3.2. Basic Scenario

1. The Business Application asks for the delivery status of a message given by a message ID.
2. The Endpoint authorizes and validates the request.
3. The Endpoint looks for the message with the given message ID in the database.
4. The Endpoint looks for the acknowledgments related to the message ID in the database.
5. The Endpoint elaborates the delivery status of the message.
6. It returns the response to the requesting Business Application.

4.3.3. Alternative Scenario 1

The message is not found in the database.

1. The basic scenario is followed until step 3.
2. The Endpoint fails to find the message in the database.
3. The Endpoint generates an error result.
4. It returns the response to the Business Application.

4.3.4. Alternative Scenario 2

The Business Application's request for validation fails.

1. The basic scenario is followed until step 2.
2. The request validation fails.
3. The Endpoint generates an error result.
4. It returns the response to the Business Application.

4.4. UC Connectivity Test

A sender Endpoint sends a tracing message to a recipient Endpoint. Each ECP component, which the tracing message passes through, sent back a trace acknowledgment to the sender Endpoint. The tracing message has a message type specified by the user, which allows testing the various message paths.

4.4.1. Activity Diagram

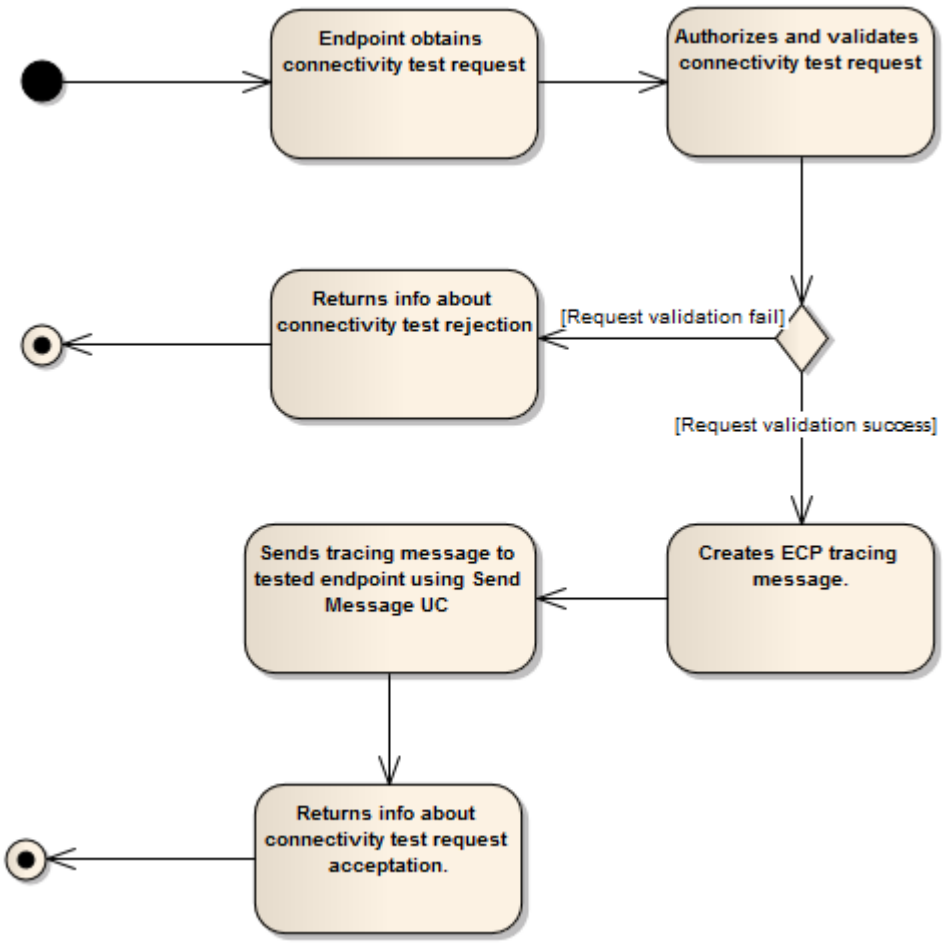


Figure 11: Connectivity test

4.4.2. Basic Scenario

1. The Business Application initiates the connectivity test (calls the Connectivity Test method on the Endpoint).
2. The Endpoint authorizes and validates the request.
3. The Endpoint creates a tracing message.
4. The Endpoint sends the tracing message to a requested recipient Endpoint using the "Send Message" use case. The user specifies the message type of the tracing message.

5. The Endpoint returns to the Business Application information about the acceptation of the connectivity test request.
6. Each component on the delivery path processes the tracing message and sends back a tracing acknowledgment to the sender Endpoint.
7. The connectivity test result is available by using the "Check Message Delivery Status" use case.

4.4.3. Alternative Scenario 1

The request validation fails.

1. The basic scenario is followed until step 2.
2. The sender Endpoint generates an error result.
3. The sender Endpoint returns the result to the requesting Business Application.

4.5. Endpoint

4.5.1. UC Registration

Each component must first register with a Component Directory (CD), referred as its home CD, to be able to exchange messages in the ECP network.

On the Endpoint, a GUI wizard supports this initialization step and leads the user through the registration process. Before the registration is completed, the Endpoint does not have a certificate set and thus messaging functionality is disabled.

The registration is a three-step process.

4.5.1.1. Import certificates

The form on the first step has a single mandatory field requiring the user to select the certificate file needed to initialize the Endpoint.

The keystore file is specific for each ECP Network and should be provided together with the ECP installation package. This file must be created by the ECP Network administrator when issuing a root certificate authority. More information is provided in section *Installation Certificate Set*.

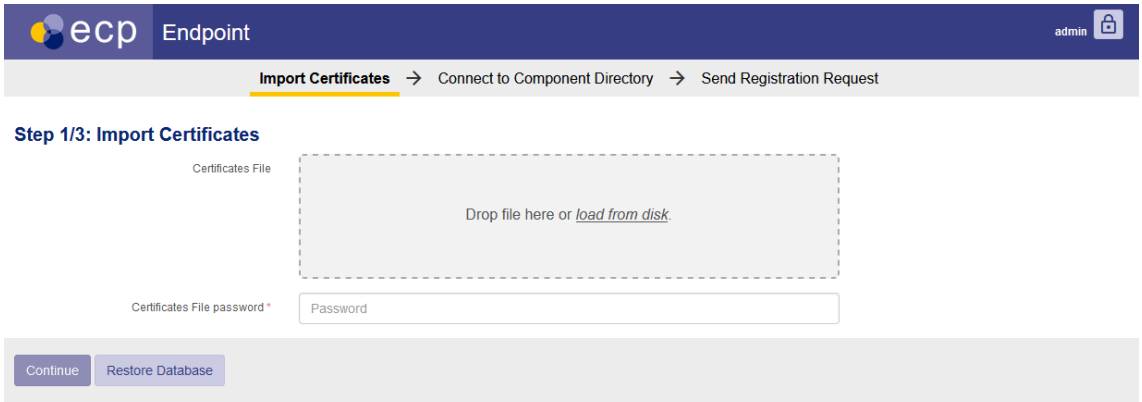


Figure 12: Import Certificates

1. The user inserts (i.e. drag and drop) the keystore file into the form. The Component Directory administrator must provide the file with the ECP installation package.
2. The user enters a password to the keystore file
3. The user clicks on the “Continue” button.
4. The Endpoint software verifies the format of the file and, if an error occurs, it is directly displayed on the page.
5. The wizard proceeds to the next step.

4.5.1.2. Database Restore

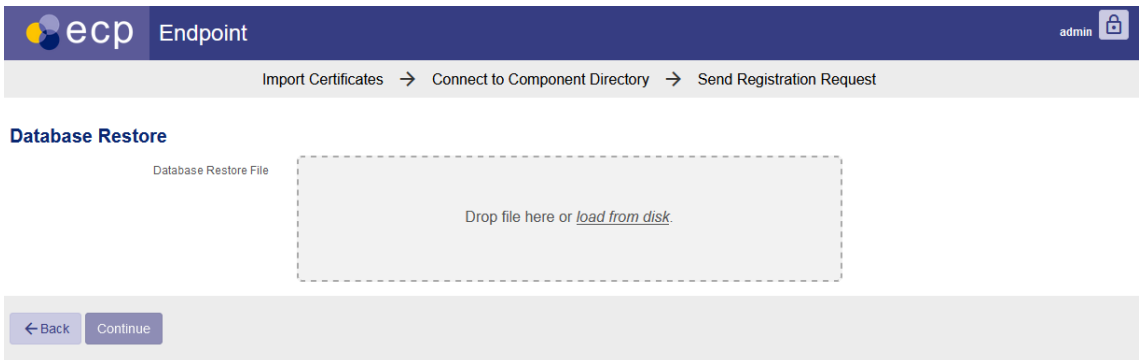


Figure 13: Database Restore

1. Instead of the proceeding with the registration, the user has an option to restore the Endpoint from a previously created database backup.
2. The user clicks on the “Restore Database” button.
3. A new page is opened, with a form to upload a database backup.
4. The user inserts (i.e. drag and drop) a database backup zip file into the form.
5. The software validates the zip file content.
   > The zip file content is described in the *Administration Guide*.

6. The software imports the database backup and replaces a content of the backed-up database tables with a content from the backup.
   > The operation is atomic and if there is an error anytime during the database restoration, no changes are saved, and the error is shown directly on the page.

4.5.1.3. Connect to Component Directory

The second step requires the user to setup the connection to the home Component Directory of the Endpoint.

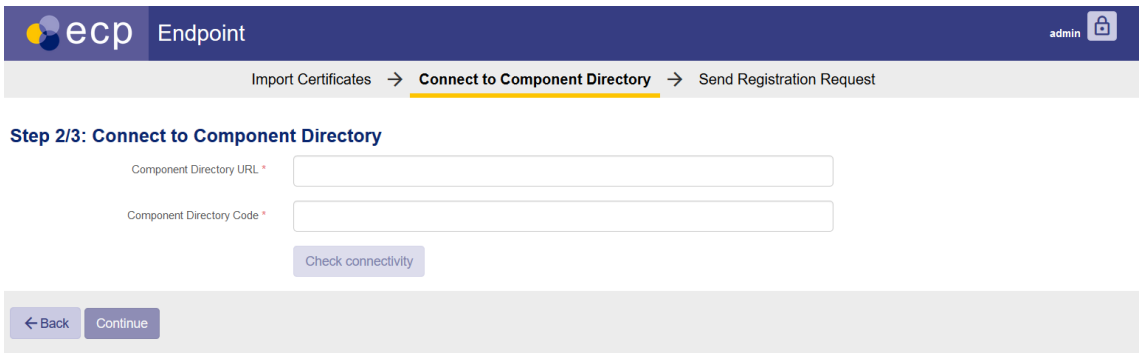


Figure 14: Connect to Component Directory

1. The user enters the URL and the code of the home (also called primary) Component Directory and clicks on the “Check connection” button.
2. The Endpoint software verifies the connectivity to the given Component Directory and shows the result on the page.
3. If the connection was successful checked, the user clicks on the “Continue” button to proceed to the next step.

Any occurring error is displayed on the screen and the registration wizard then stays on the current step. The “Continue” button is only enabled after the connection is successful checked.

4.5.1.4. Send Registration Request

The last step in the registration wizard asks the user for information about the Endpoint and about the organization where the Endpoint is installed.

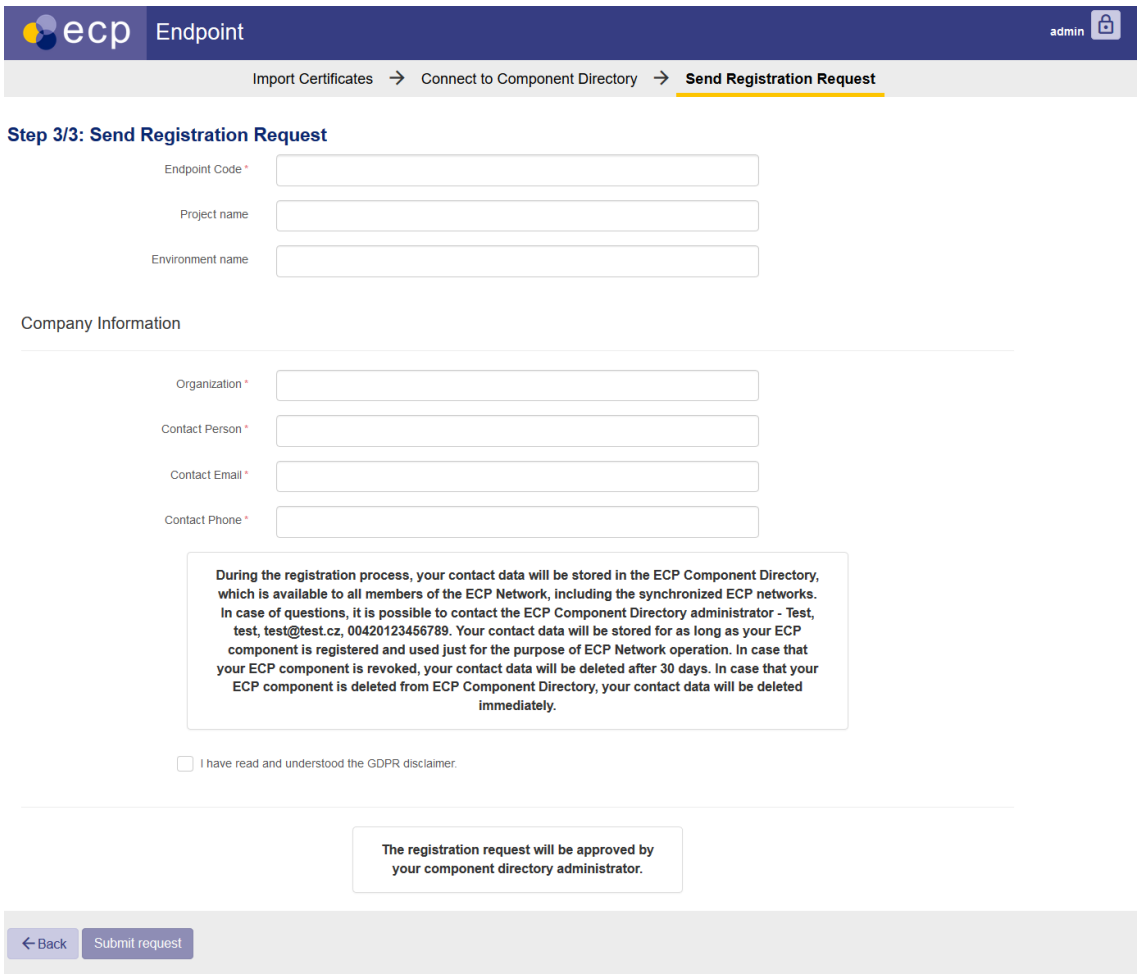


Figure 15: Send Registration Request

1. The user fills in the form, which contains the following mandatory fields:
   - Component code – code of the Endpoint
   - Organization – name of the organization owning the Endpoint
   - Contact person – name of the person responsible for the Endpoint (Endpoint administrator)
   - Contact email – email of the person responsible for the Endpoint
   - Contact phone – phone number in international format (e.g. 00<country code><number>) of the person responsible for the Endpoint
2. The user can fill in the optional fields:
   - Project name - it will be displayed in the ECP header
   - Environment name - it will be displayed in the ECP header
3. The user accepts GDPR disclaimer.

4. The user clicks on the “Submit request” button.
5. The Endpoint software validates that all data was correctly entered with the expected format.
   > If the validation fails, the Endpoint software shows the validation errors to the user and asks him to correct them before submitting the request again.
6. The Endpoint software sends the request to the home Component Directory and displays confirmation to the user. In case of connectivity error, it is displayed to the user on the returned page.

After the registration wizard is completed, the display is redirected to the Dashboard page where the user can check the registration status of the Endpoint.

4.5.2. UC Dashboard

The dashboard is a landing page for Endpoint. It provides to the user with basic information and statistics about the Endpoint behaviour. It shows following information:

- Basic Endpoint information
  > Component code
  > Description
  > Version of ECP software with optional icon and message in tooltip informing about newer software version on home Component Directory
  > Registration status (pending/active/rejected/revoked) with optional detail message in tooltip
- Total number of transported business messages from the time the Endpoint was installed
  > Total number of messages received by BA
  > Total number of messages sent by BA
- Number of waiting business messages
  > Number of messages waiting to be received by BA (i.e. incoming messages in delivered state) + a link to the Inbox page
  > Number of messages waiting to be delivered to receiver (i.e. outgoing messages in accepted state) + a link to the Outbox page
- Date and time when the last business message was
  > Sent by a business application or by a user
  > Received by a business application or by a user
- Information telling if all the Endpoint certificates are valid or not and when is the closest certificate expiration

- Information about overall message paths status (OK - green / attention - red)
  - > Number of invalid message paths with a link to the Settings page
- Information about the status of synchronization with Component Directory (OK - green / attention - red) and link to the Settings page
  - > Status with optional detail message in tooltip to provide more information about errors (e.g. last configuration push failed)
- Information about last runs of background jobs
  - > Number of completed jobs
  - > Number of jobs waiting for run
  - > Number of failed jobs
- Information about the Endpoint readiness. Endpoint is ready when it is registered, authentication certificate is not expired, it is not reloading a configuration, it is connected to an internal broker, it is synchronized with the Component Directory and there is at least one valid message path. Readiness tile is not displayed when the readiness endpoint is not enabled in the configuration properties.

4.5.2.1. GUI

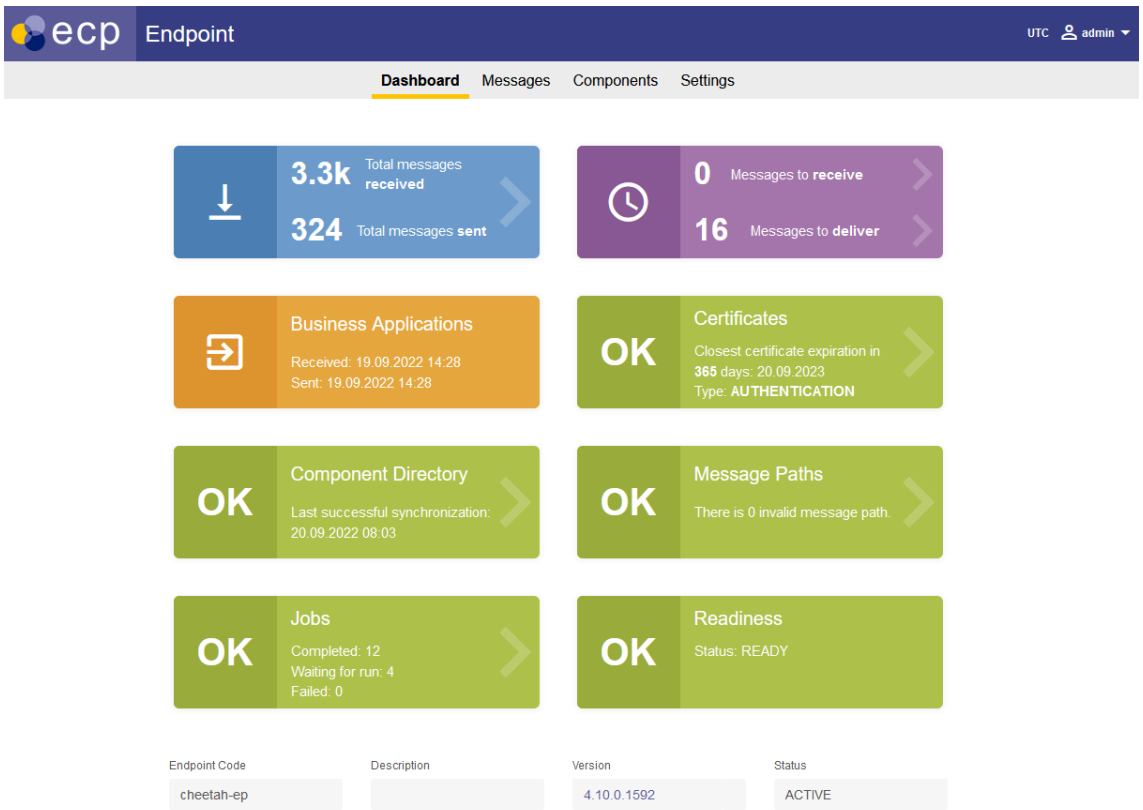


Figure 16: Endpoint - Dashboard

4.5.3. UC Message Statistics

The "Message Statistics" UC provides information about:

- The number of business messages sent or received via a Broker or an Endpoint.
- The time when the last business message was sent or received via a Broker (indirect transport) or an Endpoint (direct transport).
- If the Endpoint is currently connected to some Broker. An Endpoint can be connected to a Broker either permanently when it expects to receive business messages from it (based on the configured message paths) or temporarily when it sends a business message through the Broker and expects for the related acknowledgments to come back from that Broker. These two cases are not distinguished by this use case and are treated equally.

4.5.3.1. Basic Scenario

1. The user opens the page using the "view statistics" link on the Dashboard.
2. The "Message Statistics" page contains a table that includes all active Brokers and Endpoints that exchanged (sent or received) some business messages with this Endpoint. For each component following information is displayed:
    - Code – component code of the Broker or the Endpoint
    - Connected – if the Endpoint is currently connected to the Broker
        - In case of connectivity error, there may be optional detailed message in the tooltip
    - Sum of received messages – number of business messages received via the Endpoint from the time it was installed
    - Last message received – date time when the last business message was received from the component
    - Sum of sent messages – number of business messages sent via the Endpoint from the time it was installed
    - Last message sent – date time when the last business message was sent via the component

4.5.3.2. GUI

Message statistics

Component code	Connected	Sum message in	Last operation in	Sum messages out	Last operation out
mssql-broker	true	21.0k	10.10.2022 07:25	40.0k	19.10.2022 08:42
gorilla-broker	true	20.0k	14.10.2022 11:18	1.0k	14.10.2022 11:18
mysql-broker	false	8.0k	09.10.2022 19:39	2.0k	09.10.2022 18:12

Figure 17: Endpoint - Message Statistics

4.5.4. UC Message List - Inbox

The "Message List – Inbox" UC displays the messages delivered to the Endpoint. The user can sort and filter the list. The user can select any message in the list to display detailed information or download the content of the message into a file. In addition, the user can look for a specific message providing its message ID.

The access rights to this UC are not limited.

4.5.4.1. Basic Scenario

1. The user opens the "Message List – Inbox" page in a web browser (via the link in a menu).
2. The displayed "Message List – Inbox" page contains following information:
   - A sortable and pageable list of messages. The displayed items for each message are sender code, sender organization, message type, sending time, time of delivery to the Endpoint and message state. When Endpoint from the messages page is revoked, the organization field will be empty.
   - The content of a message in the list can be downloaded into the file using the "Download" button. The button is shown (and thus available) only if the message is not in the final state because the message content is removed immediately when the message reaches its final state.
   - The detail of a message in the list message can be displayed using the "Detail" button.
   - By default, the list is not filtered, and the items are sorted by sending time in the descending order (i.e. the last sent message is the first in the list).
   - The filter section contains the following input fields: message id, sender code with organization, message type and sending time.

4.5.4.2. Alternative Scenario – Download message content

1. The basic scenario is followed up to step 2.
2. The user starts downloading the content of the message when clicking on the "Download" button associated to the message he wants.
3. The stated of the downloaded message is marked as read, i.e. the state of the downloaded message is changed from Delivered to Received as it was received by a BA (this BA is the user itself).
4. The standard file saving dialog box is then displayed, where the proposed default file name meets the file naming conventions used to store a message in the FSSF IN directory.
5. The user chooses a destination folder and then confirms the file saving.

4.5.4.3. Alternative Scenario – Find message by Id

1. The basic scenario is followed up to step 2.
2. The user fills in the id of the message to look for and clicks on the "Search" button.
3. A page with the details of the message is displayed (otherwise the text “No information about message with id XYZ was found on this Endpoint.” is displayed).

4.5.4.4. Alternative Scenario – Download all messages

Download all provides possibility to receive all messages in Delivered state. This function is mainly used during the test phases of ECP and thus is disabled by default. It is only available when it is explicitly allowed in configuration.

1. The basic scenario is followed up to step 2.
2. The user clicks on the “Download all” button to download the content of all messages that are in the Delivered state.
3. The ECP software returns the content of all the messages in a ZIP file and moves all the messages to the Received state. The names of the files inside the ZIP archive meet the file naming conventions used to store messages in the FSSF IN directory.
4. The standard file saving dialog box is then displayed. The default file name for the zip file is *“**{EndpointCode}-inbox.zip**”.*
5. The user chooses a destination folder and then confirms the file saving.

4.5.4.5. GUI – Message List

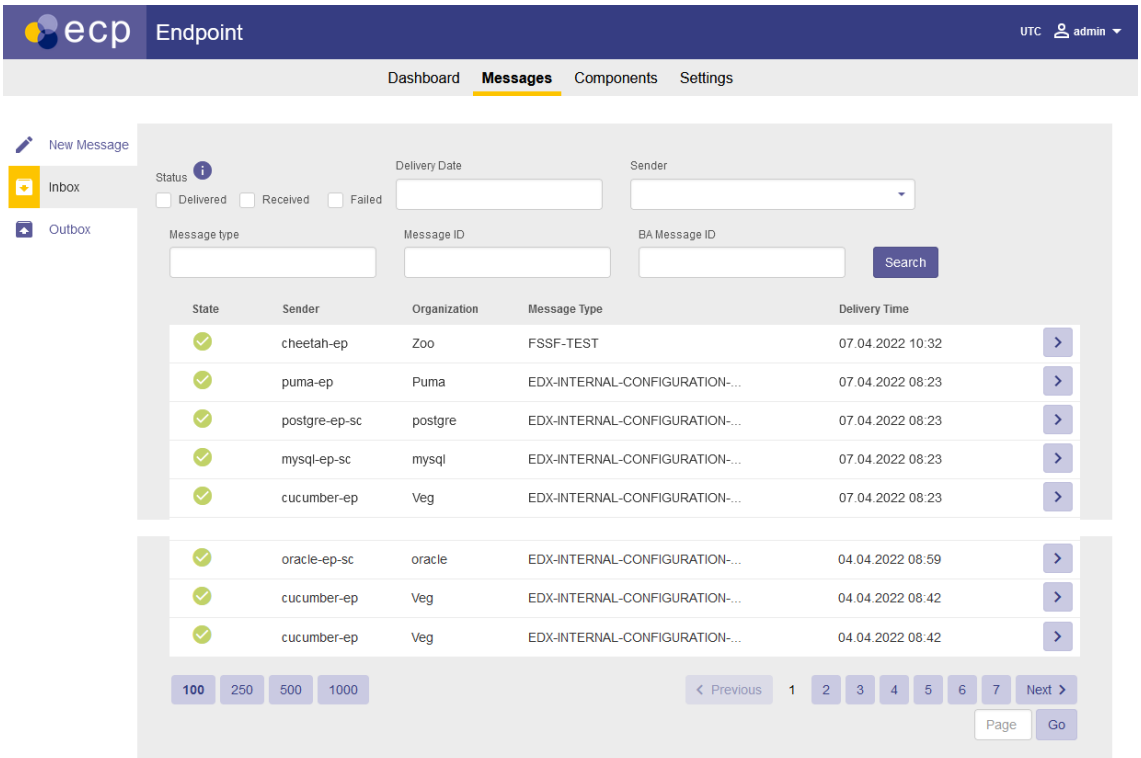


Figure 18: Endpoint - Message List - Inbox

The Message ID field together with the "Search" button allows the user to navigate directly to the detail of the message.

The filter section allows user to set the filtering criteria and to choose whether read (state Delivered), unread (state Received), failed (state Failed) or all messages should be displayed. By default, all messages are displayed, and no filtering criteria are set.

Message statuses description can be displayed in a modal window after a click on the info button in the Status filter field. Description of the specific message status is also displayed when hovering over the status icon in the State column.

The main section displays the list of messages that meet the filtering criteria. The list header is clickable, which allows changing the sorting order.

The list footer contains the paging controls (only displayed when there are at least two pages). It is possible to go to the previous page, next page, first page, last page, and several adjacent pages. It is also possible to go to the specific page by writing page number in the input field.

4.5.4.6. GUI - Message Detail

Figure 19: Endpoint - Message List - Inbox - Detail

The page contains detailed information related to the message. The Back button at the bottom allows navigating back to the list of messages and the Download button allows downloading of the message if possible (see the downloading conditions in the UC "Message List – Inbox" basic scenario).

4.5.5. UC Message List - Outbox

The "Message List – Outbox " UC displays the messages that were sent by the Endpoint. The user can sort and filter the list. The user can select any message in the list to display its detailed information. In addition, the user can look for a specific message providing its message ID.

The access rights to this UC are not limited.

4.5.5.1. Basic Scenario

1. The user opens the "Message List – Outbox" in a web browser (via a link in a menu).
2. The displayed "Message List – Outbox" page contains following information:
   - A sortable and pageable list of messages. The displayed items are receiver, receiver organization, message type, sending time and state. For each message, the Detail button is displayed. When Endpoint from the messages page is revoked, the organization field will be empty.

> By default, the list is not filtered, and the items are sorted by sending time in descending order.

> The filter section contains these input fields: receiver with organization, state, message type, sending time and message id.

4.5.5.2. Alternative Scenario – Find message by Id

1. The basic scenario is followed up to step 2.
2. The user fills in the id of the message to look for and clicks on the "Find" button.
3. A page with the details of the message is displayed (otherwise the text “No information about message with id XYZ was found on this Endpoint.” is displayed.)

4.5.5.3. GUI – Message List

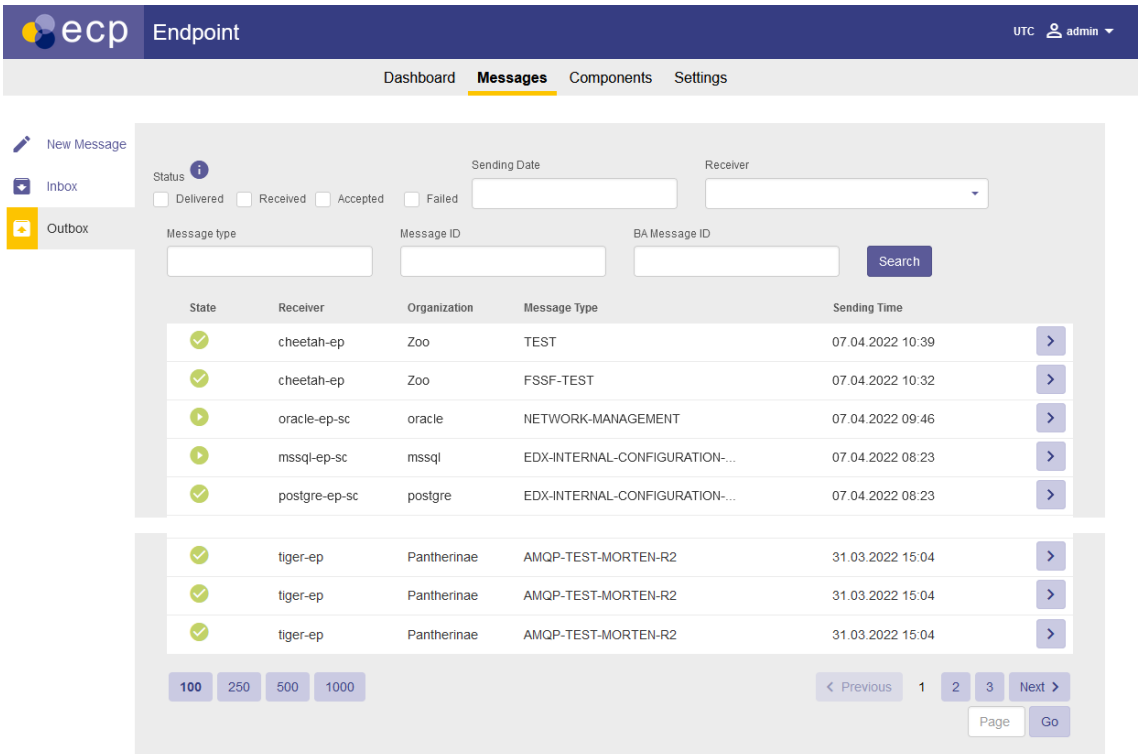


Figure 20: Endpoint - Message List - Outbox

The message ID field together with the "Search" button allows user to navigate directly to the details of the message.

The filter section allows the user to set filtering criteria. By default, all messages are displayed, and no filtering criteria are set.

Message statuses description can be displayed in a modal window after a click on the info button in the Status filter field. Description of the specific message status is also displayed when hovering over the status icon in the State column.

The main section displays the list of messages that meets the filtering criteria. The list header is clickable, which allows changing the sorting order.

The list footer contains the paging controls (only displayed when there are at least two pages). It is possible to go to the previous page, next page, first page, last page, and several adjacent pages. It is also possible to go to the specific page by writing page number in the input field.

4.5.5.4. GUI – Message Detail

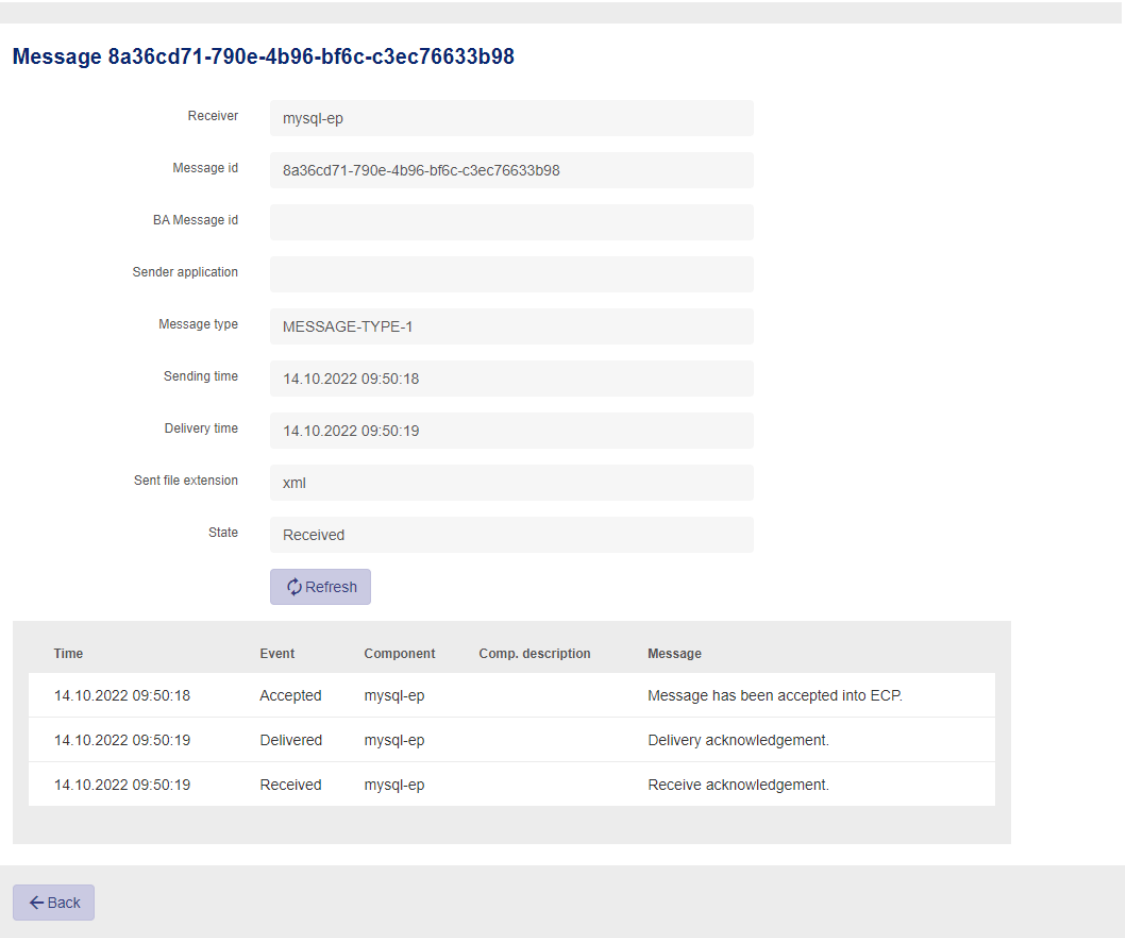


Figure 21: Endpoint - Message List - Outbox - Detail

The page contains detailed information related to the message and the details about the trace path. The Back button at the bottom allows navigating back to the list of messages and the Refresh button reloads information.

4.5.6. UC Message List - Send Message

The "Send message" UC is used to send messages through the ECP network. The access rights to this UC are not limited.

4.5.6.1. Basic Scenario

1. The user requests for sending a new message (via a link located in the menu).
2. A page with a form for sending a new message is displayed to the user. The form contains the input fields for following information: recipient Endpoint code and organization, message type, BA message ID, sender application and file to load as the content of the message.
3. The user fills in the form and submits it by clicking on the "Send" button. Only the recipient Endpoint, the message type and the content file are required. The BA message ID and the sender application are optional.
4. The message is sent to the sender Endpoint.
5. The sender Endpoint accepts the message for sending through the ECP network.
6. The outbox screen is displayed to the user together with the confirmation that the message has been accepted by the sender Endpoint.

4.5.6.2. GUI

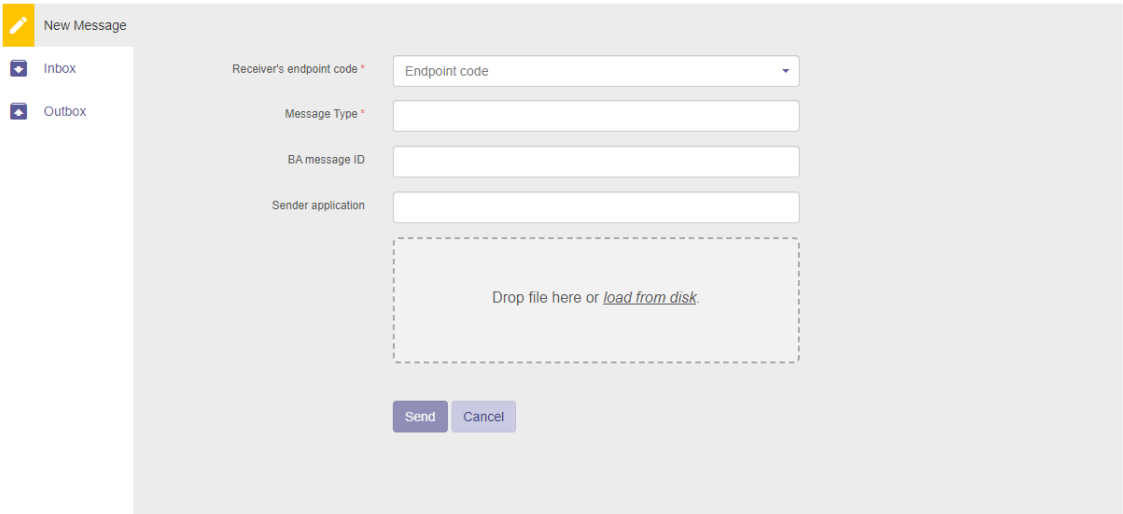


Figure 22: Endpoint - Send Message

The page contains a form for sending new message via ECP. The fields marked with an asterisk are mandatory. Two buttons are visible at the bottom of the page. The "Send" button sends the message through the ECP network, the "Cancel" button displays again the Outbox page to the user.

4.5.7. UC Components

Each Endpoint contains a "component directory browser" use case that allows the user to browse through the content of local component directory data and view the characteristics of all the active components, their message paths and their certificates. Browsing is only read-only and do not allow any modification.

The user can look at the details of a selected component or message path and see details of the component certificates. All related use cases are described in the *Component Directory* section (*Component Management*). The only difference on the Endpoint is that all use cases are read-only and do not permit any control modifying the content.

4.5.7.1. GUI

The "Components" UC on an Endpoint is a limited version of the same UC on a Component Directory. It provides a view on all the components in the ECP network, their message paths, certificate, and details.

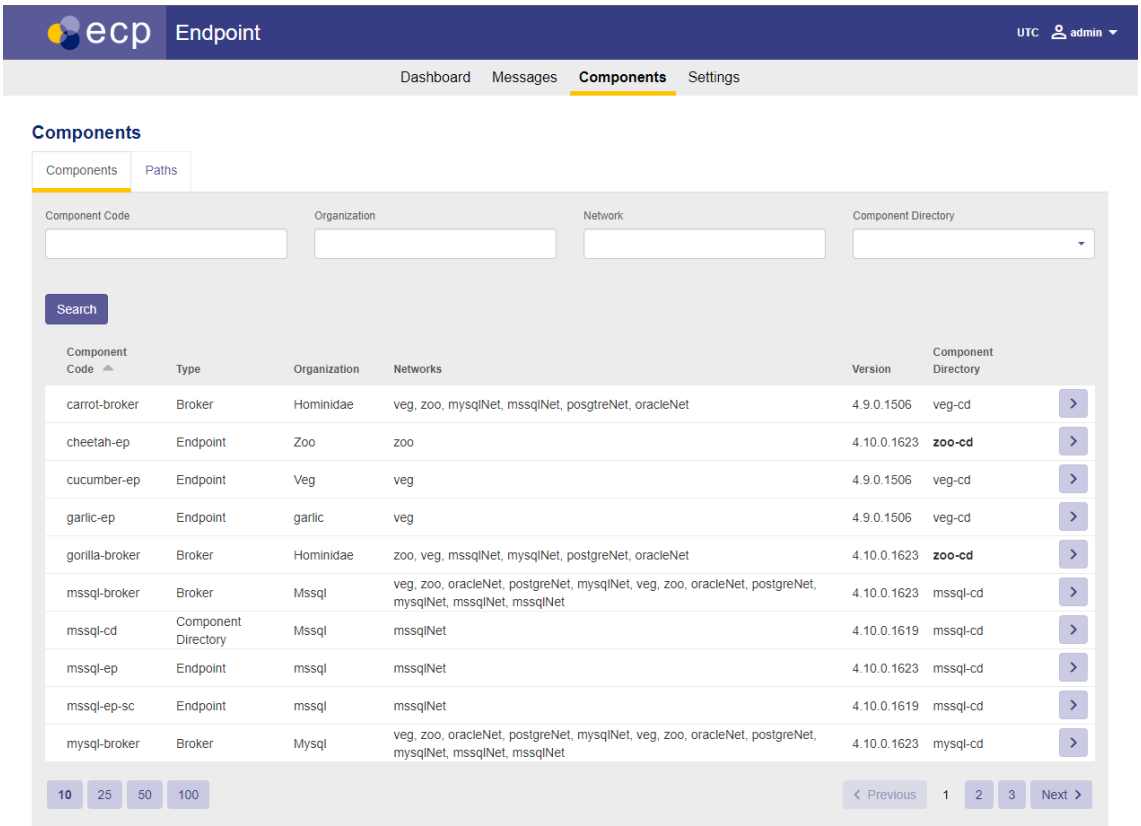


Figure 23: GUI Components

The components table shows all the components in the network with the possibility to navigate to their detail page. In case that component have not pushed its configuration yet, component is greyed out without the possibility to navigate to detail page. The list does not provide any filtering options, but sorting is possible.

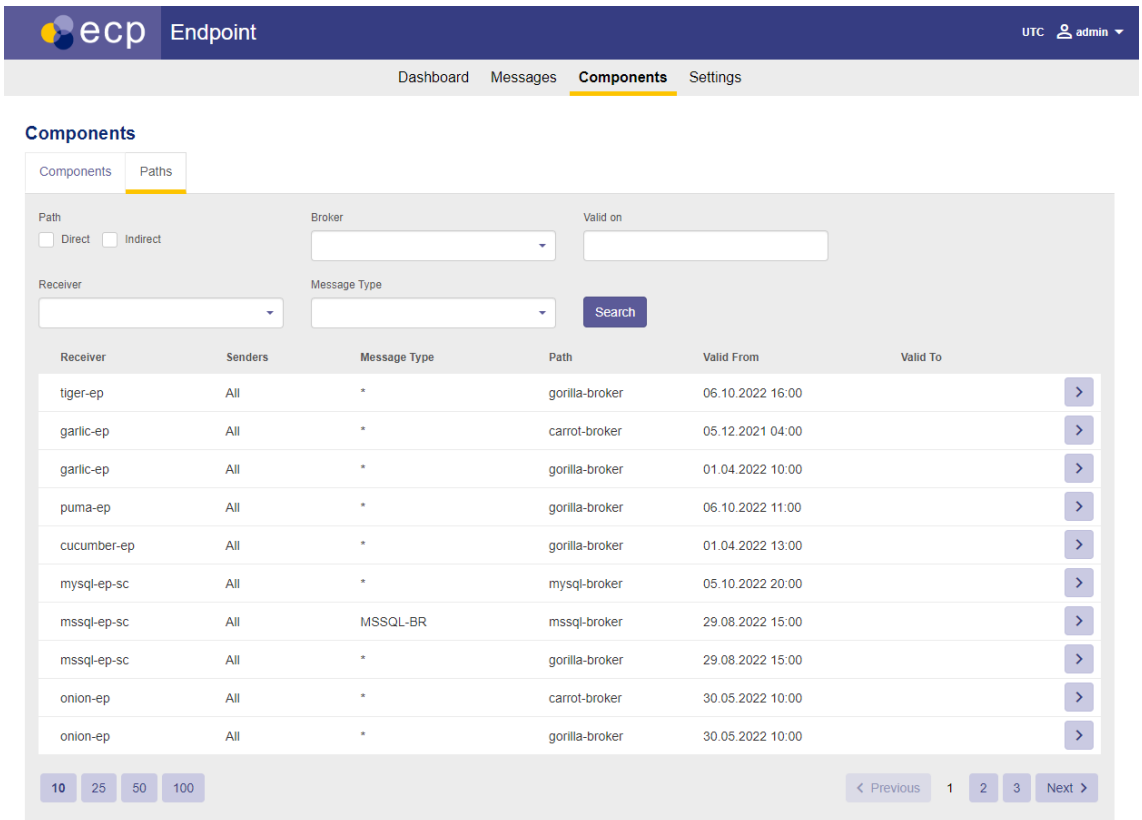


Figure 24: GUI Paths

The "message paths" table shows the defined messages paths of all the components in the network with the possibility to navigate to their detail. The table allows searching a message type to find the desired ones.

4.5.8. UC Settings

The "Endpoint settings" UC allows the user to manage the Endpoint message paths, the Endpoint certificates and the location of the component directories used by the Endpoint to synchronize.

The access rights to this UC are not limited.

4.5.8.1. Basic Scenario

1. The user opens the "Settings" page in a web browser (via a link in a menu).
2. The displayed "Settings" page contains following information:
   - Names – can be switched to editable mode to change the values
     - Project name and Environment name

- Company information – can be switched to editable mode to change the values
  - The company name and contact information (person, e-mail and phone)
- The message paths for sending messages to the Endpoint
  - Button “Create Message Path”
  - Sortable and pageable list of message paths. The displayed items are message type, path, status (i.e. active or invalid), valid from and valid to. By default, only the active paths are displayed. For each message, a "Detail" button is displayed.
  - The status may contain some text message in the tooltip to explain the invalid status, when appropriate.
- Certificates
  - Switch which enables the automatic certificate renewal
  - Button for the manual certificates’ renewal
  - Button for importing certificates from a file (typical use is update of the root certificate authority)
  - List of all the Endpoints certificates with links to their detail pages
- Component Directories
  - Configured primary and secondary component directory
  - For each component directory its code and URLs are displayed
  - Button to “Check connection to component directory”
  - Status of the synchronization with optional details in the tooltip, and the time of the last successful synchronization.
- Runtime Configuration
  - Button to export the current Endpoint configuration to a file
  - Button to import a new Endpoint configuration from a file
  - Button to edit the current Endpoint configuration in GUI
- Database Backup
  - Button to create a database backup
  - Button to restore from a database backup
- Time Zone
  - Select box for configuration of time zone
  - Save button for confirming change of time zone
- Background Jobs

- Job – background job name
- Status – job can be in the following states:
  - Completed – the last run was finished successfully
  - Waiting for run – job has not yet started
  - Failed – there was an exception in the last run
  - Disabled – job is not enabled in the configuration properties
- Start – time when the last run started
- End – time when the last run finished
- Duration – difference between start and end time
- Failure – error message, only displayed when a job is in the Failed status. Only a beginning of the message is displayed, the whole message is displayed in a modal window after a click on a detail button.

> Registration

- Button “Initiate Registration” for deleting all data and reinitialize the registration process.

4.5.8.2. Alternative Scenario – Change Automatic Certificate Renewal

1. The basic scenario is followed up to step 2.
2. The user clicks on switch to turn on/off automatic certificate renewal (chapter *Component Registration Management* explains when automatic certificate renewal is possible).
3. The software displays confirmation dialog.
4. The user confirms the action.
5. The software changes an internal parameter to the value "automatic certificate renewal"

4.5.8.3. Alternative Scenario – Component Directory Connectivity Test

1. The basic scenario is followed up to step 2.
2. The user clicks on the "Check connectivity" button for the primary or the secondary component directory.
   > Both primary and secondary URLs can be changed through the ECP configuration section on the Settings UC.
3. The software tries to connect to the selected component directory.
4. The software displays, next to the button, the result of the connection check.

4.5.8.4. Alternative Scenario – Import Certificates

The Administrator imports a certificate into the local certificate store. This is typically used to insert a new root certificate. The imported certificates are not part of the directory content.

1. The basic scenario is followed up to step 2.
2. The user clicks on the “Import certificates” button.
3. The software opens the file input dialog box and asks the user to select a file for importation.
4. The user inserts the keystore file to import and confirms the selection. The file should only be a JKS (Java Key Store) file.
5. The software validates the correct format of the file and its content.
6. The software imports the certificates contained in the file into the local certificate store. The type of certificate is determined from the certificate alias in the keystore file. More information about the keystore file can be found in chapter *Installation Certificate Set*.

4.5.8.5. Alternative Scenario – Import Configuration

1. The basic scenario is followed up to step 2.
2. The user clicks on the “Import Configuration” button.
3. The software opens the file input dialog box and asks the user to select a file to import.
4. The user selects the configuration file to import and confirms the selection.
5. The software validates the correct format of the file and its content.
   - The format and all configuration parameters are described in the *Administration Guide*.
6. The software imports the configuration into the local database and updates the running application.
   - The operation is atomic and if there is an error anytime during the import, no changes are saved, and the error is shown directly on the page.

4.5.8.6. Alternative Scenario – Export Configuration

1. The basis scenario is followed up to step 2.
2. The user clicks on the “Export Configuration” button.
3. The software opens the file output dialog box and asks the user to select where to save the exported configuration file.
4. The user selects a location for export and confirms.

5. The software exports the current configuration into a file named *"**{COMPONENT-CODE}-configuration.properties**"*.

4.5.8.7. Alternative Scenario – Edit Configuration

1. The basic scenario is followed up to step 2.
2. The user clicks on the "Edit Configuration" button.
3. The software opens a new page with all runtime properties.
4. The user edits the properties and clicks on the Save button.
5. The software validates the correct format of the changed properties.
   > The format and all configuration parameters are described in the *Administration Guide*.
6. The software imports the configuration into the local database and updates the running application.
   > The operation is atomic and if there is an error anytime during the import, no changes are saved, and the error is shown directly on the page. Invalid properties are also highlighted in a red colour.

4.5.8.8. Alternative Scenario – Create Database Backup

1. The basis scenario is followed up to step 2.
2. The user clicks on the "Create Backup" button.
3. The software creates a database backup and exports it into a zip file named *"**{COMPONENT-CODE}_{current timestamp}.zip**"*.

4.5.8.9. Alternative Scenario – Restore Database Backup

4. The basis scenario is followed up to step 2.
5. The user clicks on the "Restore Backup" button.
6. The software displays a warning dialog box.
7. The user clicks on the "Proceed" button.
7. The user selects a database backup zip file to import and confirms the selection.
8. The software validates the zip file content.
   > The zip file content is described in the *Administration Guide*.
9. The software imports the database backup and replaces a content of the backed-up database tables with a content from the backup.
   > The operation is atomic and if there is an error anytime during the database restoration, no changes are saved, and the error is shown directly on the page.

4.5.8.10. Alternative Scenario – Edit Company Information

1. The basis scenario is followed up to step 2.
2. The user clicks on the “Edit” button assigned to the section with company information.
3. The software switches the company information to editable mode allowing the user to change the values.
4. The user makes the required changes and confirms them by clicking on the “Save” button.
   - Alternatively, the user may discard the changes using the “Cancel” button.
5. The software saves the changes and pushes them to the home Component Directory.

4.5.8.11. Alternative Scenario – Change Time Zone

1. The basis scenario is followed up to step 2.
2. The user selects time zone from select box.
3. The user confirms the change by clicking on the “Save” button.
4. The software saves the changes and reloads the GUI with times in the selected time zone.
   - Note that the time of already sent/receive messages is not changed. The new time zone is applied only for new messages.

4.5.8.12. Alternative Scenario – Reinitiate registration

1. The basis scenario is followed up to step 2.
2. The user clicks on the “Initiate Registration” button
3. Confirmation popup window is displayed
4. The user confirms initialization of registration process by clicking on the button “Proceed”.
   - Alternatively, the user may discard the reinitialization using button “Back”.
5. The registration screen is displayed.
   - During registration process all original registration data are prefilled and can be changed.

4.5.8.13.GUI

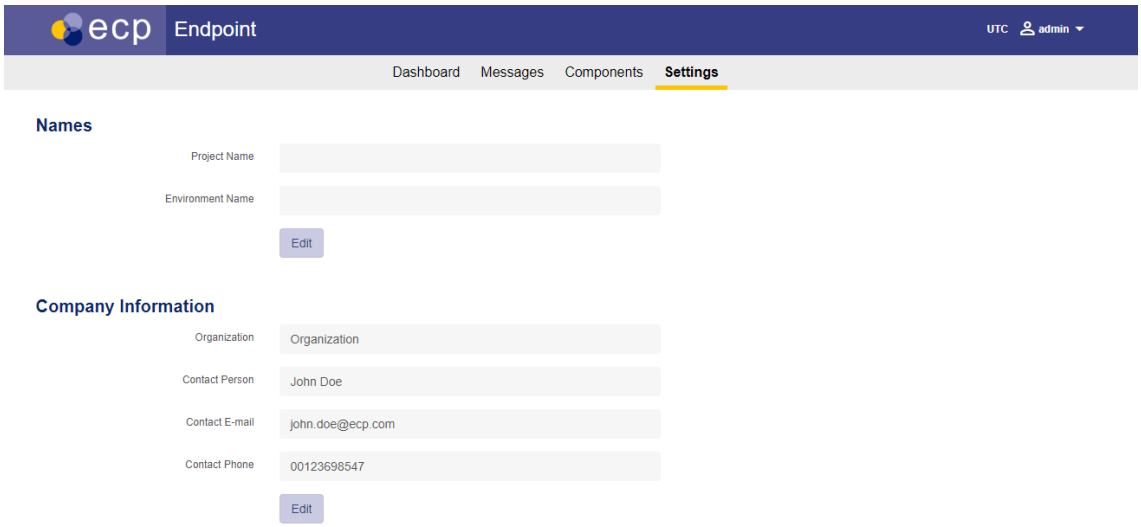


Figure 25: Endpoint - Settings, part 1

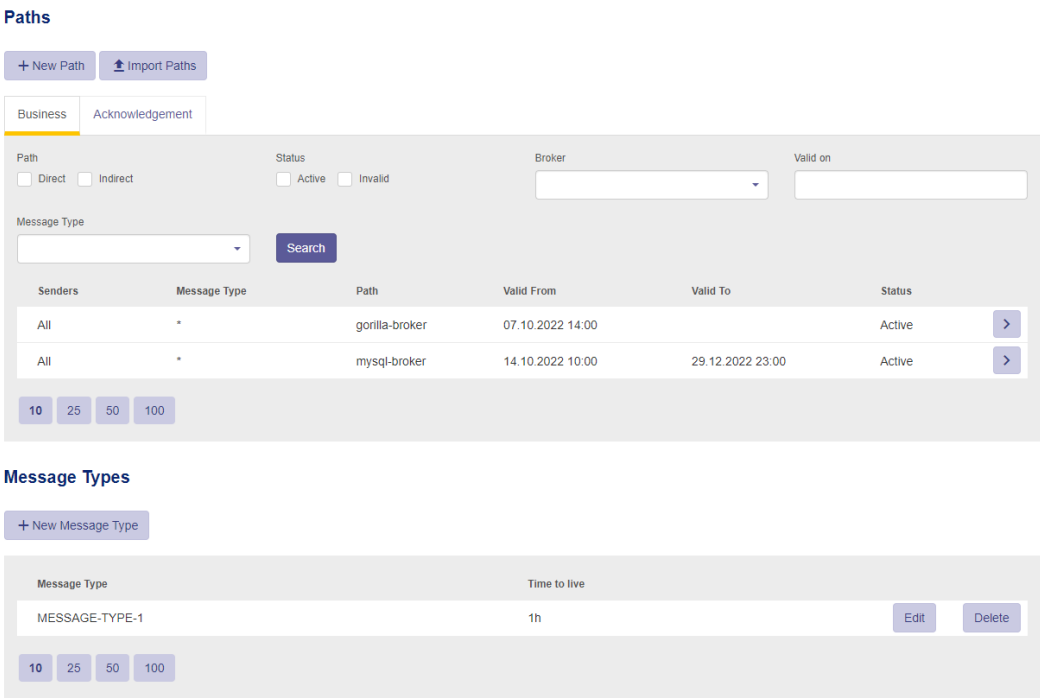


Figure 26: Endpoint - Settings, part 2

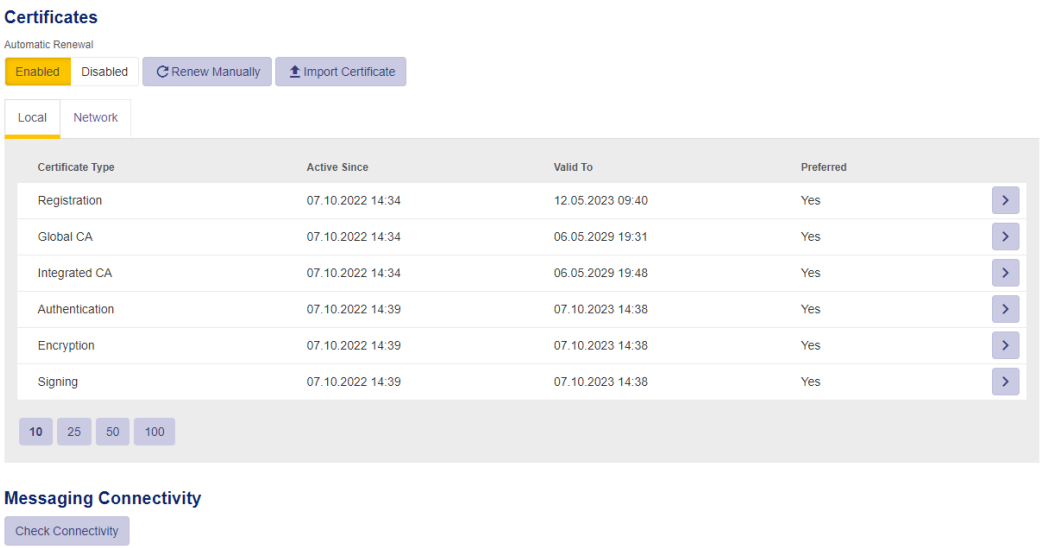


Figure 27: Endpoint - Settings, part 3

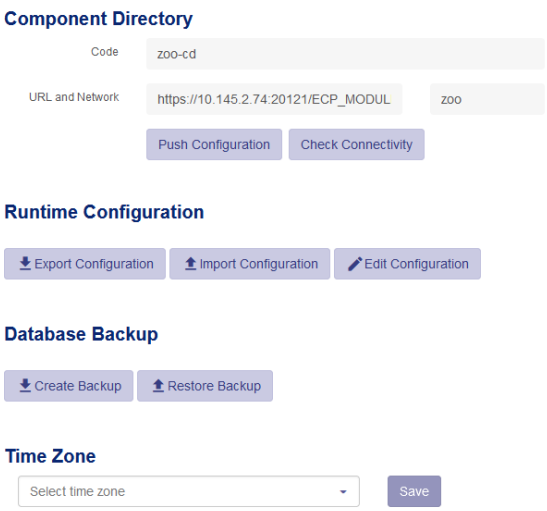


Figure 28: Endpoint - Settings, part 4

Background Jobs

Job	Status	Start	End	Duration	Failure
Cluster Status Manager	Completed	07.04.2022 06:33:26.287	07.04.2022 06:33:26.289	2 ms	
Configuration Reload	Completed	07.04.2022 06:33:25.001	07.04.2022 06:33:25.004	3 ms	
Synchronization	Completed	07.04.2022 06:30:00.006	07.04.2022 06:30:01.667	1661 ms	
Certificate Renewal	Completed	06.04.2022 09:50:00.042	06.04.2022 09:50:00.370	328 ms	
Database Compression	Completed	06.04.2022 09:30:15.090	06.04.2022 09:30:15.357	267 ms	
Expired Certificates Cleaning	Completed	06.04.2022 09:10:00.208	06.04.2022 09:10:00.336	128 ms	
Directory Cache Reload	Waiting for run				
Routes Management	Completed	07.04.2022 06:00:00.015	07.04.2022 06:00:00.125	110 ms	
ECP3 Message Download	Waiting for run				
ECP3 Cache Expiration	Waiting for run				

10 25 Previous 1 2 Next

Registration

Initiate Registration

Figure 29: Endpoint - Settings, part 5

4.5.9. UC Renew Certificates

Through the "Renew Certificates" UC, The Endpoint administrator can manually renew the certificates of the Endpoint.

4.5.9.1. Basic Scenario

1. The user opens the "Renew Certificates" UC via a link on the "Settings" page.
2. The page contains a form where the user can enter following information: Organization, contact person, contact email, and contact phone. All the fields are mandatory and prefilled to the values obtained from Endpoint configuration.
3. The user verifies and optionally changes the form values and submits the request by clicking on the “Submit request” button.
4. The Endpoint sends the request for new certificates to the home Component Directory.
5. The Endpoint redirects the user to the "Settings" page and displays a notification about the operation result. The user is informed if the request was automatically confirmed or if it waits for the manual approval of the Component Directory administrator.

4.5.9.2. GUI

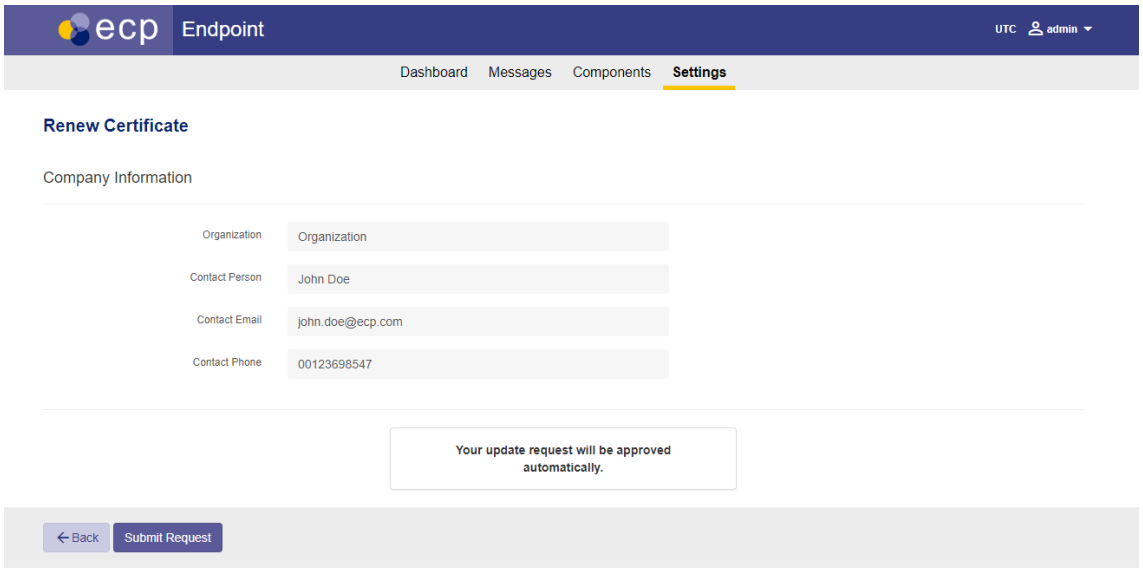


Figure 30: Update Certificates GUI

The page shows contact information for verification and possible correction. It also indicates whether the request will be approved automatically or if a manual approval of the Component Directory administrator will be necessary. The user submits the request for new certificates using the “Submit request” button.

4.5.10. UC Certificate Detail

The "Certificate Detail" page shows information about the user selected certificate and may provide buttons for changing the certificate state.

Following information are displayed for the certificate:

- The validity period of the certificate
- The type of the certificate, i.e. its usage within ECP (Authentication, Signing or Encryption)
- The current status of the certificate (active, expired, revoked)
- The name of the certificate authority that signed the certificate
- Information if the certificate is currently preferred (default) for given certificate type. In case there are multiple certificates of the same type valid at one time, the preferred certificate is used in corresponding security operations.
  - > The preferred certificate is selected automatically by the system, and it is the oldest valid certificate of given type. This is to enable the synchronization process to distribute new certificate to all ECP components. A renewed certificate replaces the older when it expires.

If the certificate is owned by the user's Endpoint, a "Delete" button is available allowing the user to remove the certificate and corresponding private key from the Endpoint.

4.5.10.1. GUI

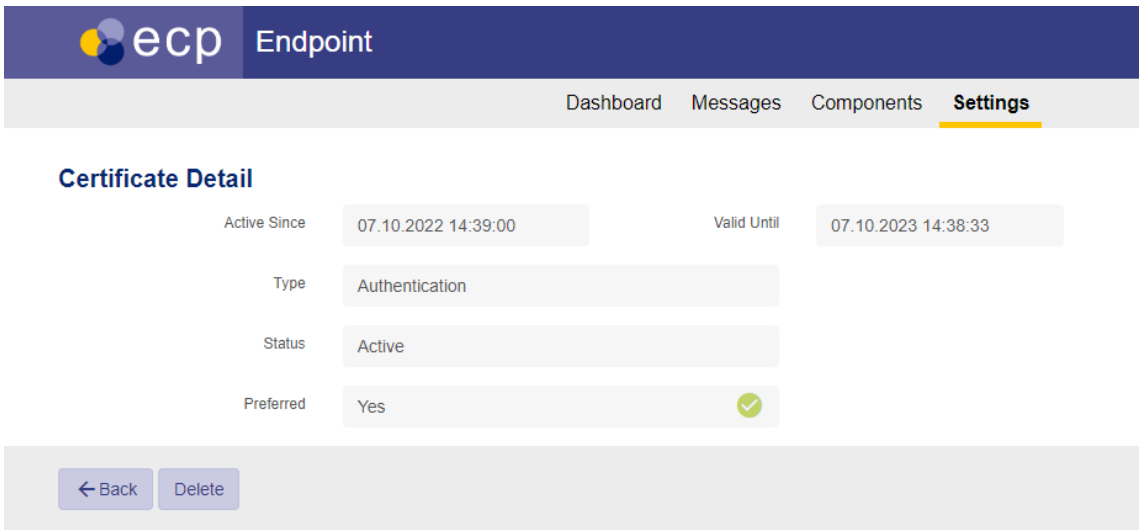


Figure 31: Certificate Detail GUI

4.5.11. Message Path Management

A message path defines how a message is transferred to an Endpoint. Each Endpoint administrator can only configure the receiving message paths, i.e. how the messages are transferred or delivered to his Endpoint. Such defined message paths are used to route the business messages.

The acknowledgments always follow the same route of their original message (but in the reverse way).

The attributes of a message path are:

- Sender –The codes (one or more) of the sender Endpoints which messages can be transferred using the message path. A wildcard (*) means any sender
- Message type – The messages having this message type can be transferred using the message path. The message type can contain a wildcard (*) at the end (e.g.: ABC*)
- Path – information about transport type for the message path: Direct or indirect through a Broker. ECP v4 supports only one Broker in a message path and therefore does not allow a message to be routed through multiple Brokers.

- From – the date time (inclusive) from which the message path can be used to send messages. It can be specified with an hourly precision.
- To – the date time (exclusive) until the message path is valid. After this time, any sender should not use the message path. This parameter is optional. When not set, it means that the message path is forever valid. It can be specified with an hourly precision.

It is not possible to create two message paths with the same message type and with overlapping validity intervals (from – to). This means, on the contrary, that there may exists message paths with the same message type but with non-overlapping validity intervals.

Example:

- The following situation is correct.
  - Message path 1: message type = TypeA from = 1.1.2016 00:00 to 1.7.2016 00:00
  - Message path 2: message type = TypeA from = 1.7.2016 00:00 to 1.8.2016 00:00

Please note that when a message type ends with the wildcard character, the message path does not conflict with a specific message type. This means that there may exists a message path with message type = "TypeA" and another message path with message type = "TypeA*" both with the same or overlapping validity intervals.

The configured message paths are used to route the messages in the ECP network. This means that, if there is no configured message path for a given message type and a recipient Endpoint, then the message will not be sent and will be moved to a failed state. It is then recommended that every Endpoint administrator defines a so-called “default message path”, which message type is “*”.

4.5.11.1. Message Paths for the Acknowledgements

The acknowledgements do respect the message paths defined by the recipient Endpoint of the original message. They always use the same path as their original business message. This means that if an original message has been sent using the indirect communication pattern via some Broker, then the corresponding acknowledgement will be sent using the same Broker. If the original business message has been sent using the direct communication pattern, then the acknowledgement will be also sent using the direct communication pattern.

Therefore, a sender Endpoint that does not accept incoming direct communication (e.g. for security reason) never accept a message to send by using direct communication, for no acknowledgement back is possible.

Because of this behaviour, it is very important, when defining a message path using direct communication that the administrator checks that all expected sender Endpoints also accept incoming direct communication.

Acknowledgment message paths are displayed in the Message Paths table in the Setting page of Endpoint GUI. This table is divided into two tabs – Business for standard message paths and Acknowledgment for acknowledgment message paths. Detail page of an acknowledgment message path can be displayed. Acknowledgment message path can also be deleted from the detail page because acknowledgment message paths are not removed automatically.

4.5.11.2. Finding a Valid Message Path

When the sender Endpoint is sending message, it must find the message path to use to transport the message through the network. The algorithm is as follows:

1. Get all declared message paths for the recipient Endpoint.
2. Filter out all message paths having the current time out of the validity period.
3. Find the message path with the most specific message type. Following rules are used:
   - Equal message type is the most specific.
   - Message type plus wildcard is the second most specif.
   - Message type substring plus wildcard is the third most specific. Please note that the algorithm will take the longest matching substring.
   - Wildcard only is the least specific.

When the message path is found, the software validates if the message sender is allowed by the message path and if the intermediate Broker (for indirect communication) or Endpoint (for direct communication) is attached to the same physical network as the sending Endpoint (see the System Design for more information on networks management). If the message sender is not allowed, then it is not possible to send messages with that message type to the recipient Endpoint.

Example:

- Endpoint 1 declares the following message paths.
  - Message path 1: message type = Type*, from = 1.1.2016, to = infinity, sender = all
  - Message path 2: message type = TypeA, from = 1.1.2016, to 31.1. 2016, sender = all
  - Message path 3: message type = TypeA*, from = 1.2.2016, to 10.2. 2016, sender = Endpoint 3

Let us assume that on 2.1.2016 Endpoint 2 is sending a message with message type = TypeA to Endpoint 1. In the step 2, the software will filter out message path 3, because it is not valid for the current time. In the step 3, the software will choose message path 2 because its message type is the most specific one.

Let us assume that on 2.1.2016 Endpoint 2 is sending a message with message type = TypeC to Endpoint 1. In the step 2, the software will filter out message path 3, because it is not valid for the current time. In the step 3, the software will choose message path 1 because its message type is the most specific one.

Let us assume that on 1.2.2016 Endpoint 2 is sending a message with message type = TypeA to Endpoint 1. In the step 2, the software will filter out message path 2, because it is not valid for the current time. In the step 3, the software will choose message path 3 because its message type is the most specific one. However, in this situation Endpoint 2 is not able to send message to Endpoint 1 because the message path limits the senders only to Endpoint 3.

4.5.11.3. Conflicts with External Rules

It may happen that already existing message paths become conflicting with new rules of other ECP components. Typical example of these rules are new Broker restrictions (see section *Restricting Message Paths*), which can prevent messages of selected message types from being sent through that Broker.

If the Broker administrator defines a new restriction that forbids TypeX from being transferred through that Broker and there are message paths defined using that Broker with affected message type, then they will become invalid.

A message Path may become invalid when:

- It uses an intermediate Broker that was revoked or removed from the Component Directory
- The path restrictions of the intermediate Broker do not allow message type or receiver of the message path
- The intermediate Broker is no longer attached to the same physical network as the Endpoint

There are some consequences when a message path becomes invalid due to conflict with other rules:

- The messages routed through the affected path that are still on the sender will become undeliverable and will be moved to the failed state.
- The messages that are already on the Broker or on the receiver will be delivered successfully, but the recipient Endpoint will not be able to send back the acknowledgment through the Broker.
    - As the acknowledgments are always sent back through the same path from which the original message was received, they will become undeliverable and will be moved to failed state.
- The administrator of an Endpoint with conflicting message paths receives a notification about the number of conflicting message paths on the Dashboard. The administrator uses filter on the message paths list to quickly find conflicts and resolve them.

4.5.12. UC Create Message Path

The Endpoint administrator can create a new message path to his Endpoint using this use case.

4.5.12.1. Basic Scenario

The basic successful scenario.

1. The user opens the web page using the "new path" button on the "Settings" page.
2. The user fills the form and submits the change using the "save" button. The values available on the form are senders, message type, path, valid from and valid to. All the fields are mandatory except for "valid to", which can be left empty to not limit the end of the validity period.
3. The software performs the validations for the mandatory fields and the value format.
4. The software verifies that there are no collisions with other defined routing rules.

4.5.12.2. It does not conflict with external rules (see Finding a Valid Message Path)

When the sender Endpoint is sending message, it must find the message path to use to transport the message through the network. The algorithm is as follows:

1. Get all declared message paths for the recipient Endpoint.
2. Filter out all message paths having the current time out of the validity period.
3. Find the message path with the most specific message type. Following rules are used:
   - Equal message type is the most specific.
   - Message type plus wildcard is the second most specif.
   - Message type substring plus wildcard is the third most specific. Please note that the algorithm will take the longest matching substring.
   - Wildcard only is the least specific.

When the message path is found, the software validates if the message sender is allowed by the message path. If the message sender is not allowed, then it is not possible to send messages with that message type to the recipient Endpoint.

Example:

- Endpoint 1 declares the following message paths:
  - Message path 1: message type = Type*, from = 1.1.2016, to = infinity, sender = all
  - Message path 2: message type = TypeA, from = 1.1.2016, to 31.1. 2016, sender = all

> Message path 3: message type = TypeA*, from = 1.2.2016, to 10.2. 2016, sender = Endpoint 3

Let us assume that on 2.1.2016 Endpoint 2 is sending a message with message type = TypeA to Endpoint 1. In the step 2, the software will filter out message path 3, because it is not valid for the current time. In the step 3, the software will choose message path 2 because its message type is the most specific one.

Let us assume that on 2.1.2016 Endpoint 2 is sending a message with message type = TypeC to Endpoint 1. In the step 2, the software will filter out message path 3, because it is not valid for the current time. In the step 3, the software will choose message path 1 because its message type is the most specific one.

Let us assume that on 1.2.2016 Endpoint 2 is sending a message with message type = TypeA to Endpoint 1. In the step 2, the software will filter out message path 2, because it is not valid for the current time. In the step 3, the software will choose message path 3 because its message type is the most specific one. However, in this situation Endpoint 2 is not able to send message to Endpoint 1 because the message path limits the senders only to Endpoint 3.

4. Conflicts with External Rules
   > It does not conflict with other message paths (having the same message type and an overlapping validity period)
5. The software stores the message path to the local Component Directory.
6. The software pushes the change in message paths to home Component Directory.

Note that the order in which the message path is saved is important: the message path is first stored locally (in a separate transaction) and only then, it is synchronously pushed to the home Component Directory. This principle is described in the "System Design" document together with the process of pushing component configuration data to the home Component Directory.

4.5.12.3. Alternative Scenario – Path Validation Failed

The validation of the path failed because some mandatory fields were not filled or because some values do not comply with their required format.

1. The basic scenario is followed up to point 3.
2. The software detects a validation error, displays the message to the user and highlights the incorrect values.

4.5.12.4. Alternative Scenario – Path Collision Detected

The validation failed because the path collides with another routing rule.

1. The basic scenario is followed up to point 4.
2. The software detects a collision with another rule.

3. The software displays a validation error with details about the message path that caused the collision.

4.5.12.5. Create Message Path – GUI

The form contains following fields:

- Senders – The default value is “All”.
- Message type – The text field with wildcard (“*”) as the default value
- Path – direct or indirect. In case indirect is selected, additional code of Broker is required.
- Valid from – the date and time from which the message path can be used in messaging. The default is the current date and time rounded down to the current hour.
- Valid to – date and time until which the message path can be used in messaging. Field is optional and by default is empty meaning there is no validity end.

Screenshot with “All” senders selected:

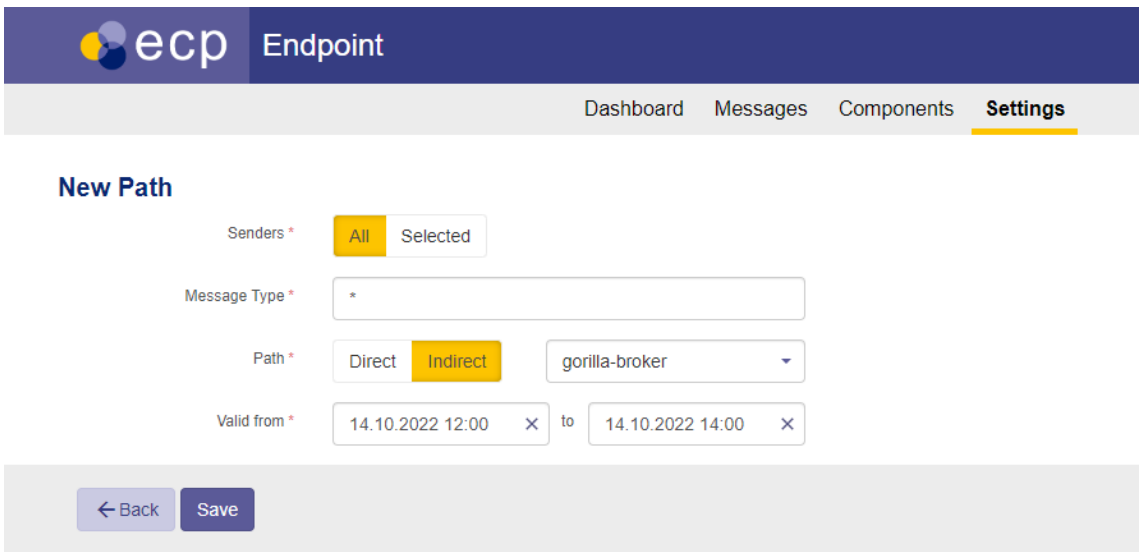


Figure 32: New Path GUI

Screenshot with selection of senders:

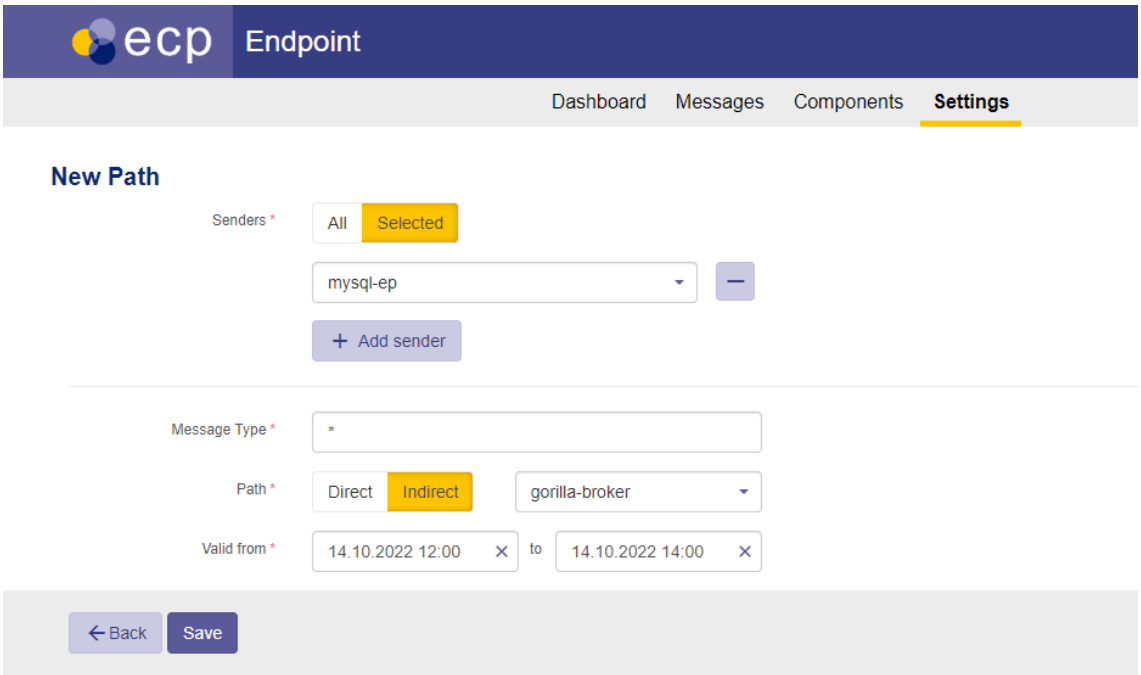


Figure 33: New Path GUI 2

The user can select multiple senders by the using “Add sender” button.

4.5.13. UC Message Path Detail

The "Message path detail" shows all information about a message path.

4.5.13.1. Basic Scenario

The basic successful scenario.

1. The user opens the web page using a “Detail” button on the "Settings" page.
2. The details about the message path are displayed.

4.5.13.2. Alternative Scenario – Delete Path

1. The basic Scenario is followed to the end.
2. The user clicks on the “Delete path” button.
3. The software asks for confirmation that selected path should be deleted.
4. The user confirms the action.
5. The system removes the message path from directory.

4.5.13.3. GUI

The page contains important information about message path.

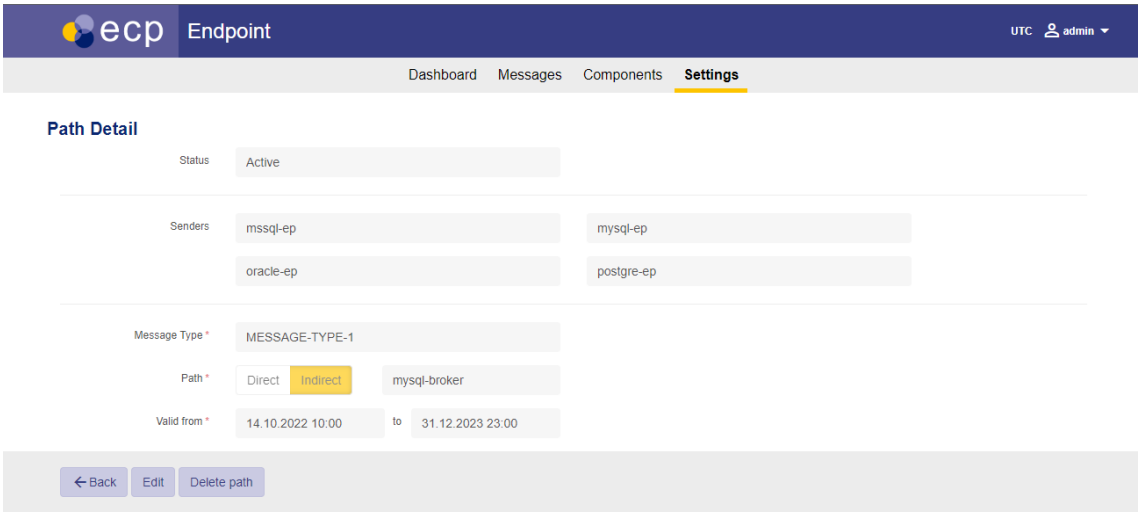


Figure 34: Path Detail GUI

4.5.14. UC Edit Message Path

The Endpoint administrator can edit, i.e. modify, an existing message path using this use case.

4.5.14.1. Basic Scenario

The basic successful scenario.

1. The user opens the web page using the edit button in the message path detail page.
2. The edit page is loaded with current values for the selected message path.
3. The user fills the form and submits the change using the save button. Values editable on the form are senders, valid from and valid to. All the fields but “valid to” are mandatory.
   - To adjust non-editable fields, administrator must set “valid to” attribute to existing field to set the end of its validity and create a new message path with “valid from” equal to “valid to” of the existing message path. This ensures continuity of messaging.
4. The software performs validations for mandatory fields and value format.

5. The software verifies there are no collisions with other and already existing routing rules (message paths, Broker restrictions).
6. The software stores the message path to the local Component Directory.
7. The software pushes the change in message paths to home Component Directory.

4.5.14.2. Alternative Scenario – Path Validation Failed

Validation of the path failed because some mandatory fields were not filled, or values do not comply with the required format.

1. The basic scenario is followed up to point 4.
2. The software detects a validation error, displays the message to the user and highlights the incorrect values.

4.5.14.3. Alternative Scenario – Path Collision Detected

Validation failed because the path collides with another routing rule.

1. The basic scenario is followed up to point 5.
2. The software detects a collision with another existing message path.
3. The software displays a validation error with details about the message path that caused the collision.

4.5.14.4. Edit Message Path – GUI

The form contains following fields, which are set to the current values:

- Senders – Allowed sender using this message path.
- Valid from – date and time from which the message path can be used in messaging.
- Valid to – date and time until which the message path can be used in messaging. Field is optional.

Wireframe with selection of senders:

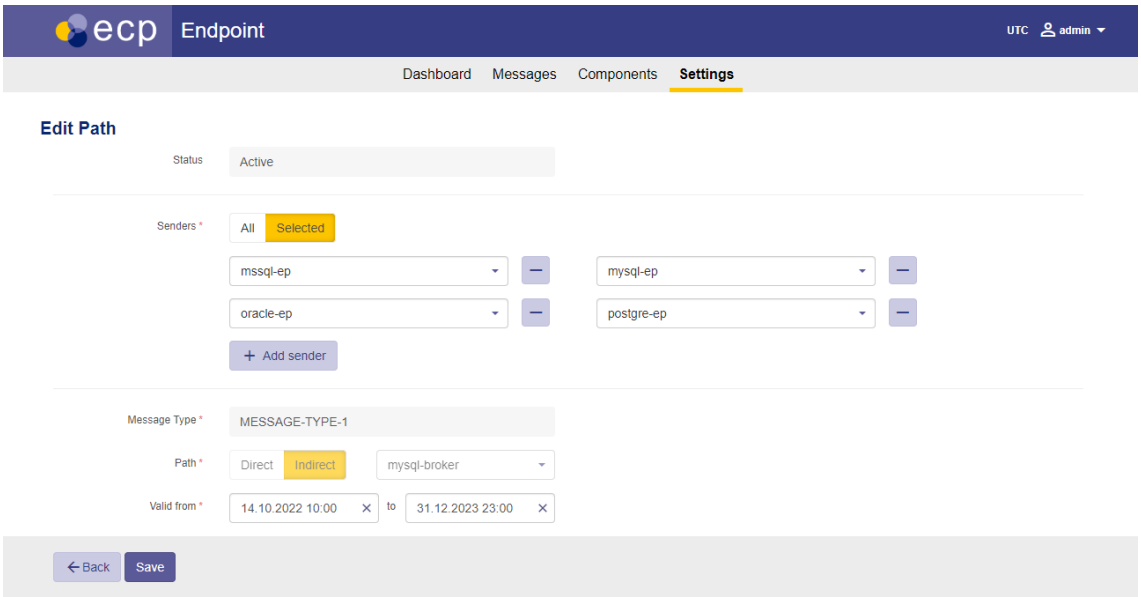


Figure 35: Edit Path GUI

4.6. Component Directory

Component Directory is a standalone component containing key information about all the components registered into the ECP network, which are Endpoints, Brokers, and other Component Directories.

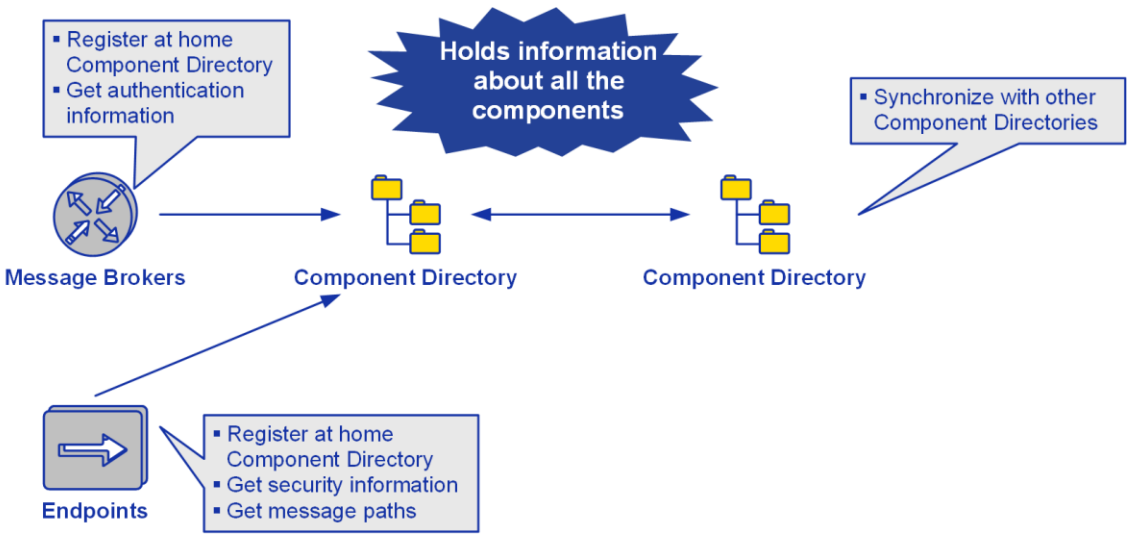


Figure 36: ECP High-level architecture - Component Directory

The key responsibility and features of the Component Directory are:

- Provide registration interface to the Endpoints and the Brokers, which use it to request new registration within their home Component Directory or renew certificates for an already registered component.
- Issue the certificates used by the ECP components.
- Maintain security information (signing, encryption, and authentication certificates) about all the Endpoints in the network
- Maintain and provide the message paths established between the Endpoints
- Mutually synchronize the Directory content with other Component Directories to maintain consistent information throughout the ECP network
- Provide the synchronization interface which is used by other ECP components

4.6.1. Component Registration Process

Each Endpoint/Broker must register with a Component Directory, referred as its home Component Directory. This is realised by the component registration process. At the beginning of the process, the ECP network does not know the component. At the end of the process, the component is registered, and it may use all ECP functions. This registration process consists of several automatic/manual steps.

1. Endpoint/Broker
   - Configure the home component directory.
   - Send request to the home Component Directory to issue certificates. The request contains:
     - Required component code
     - Organization
     - Contact Person
     - Email
     - Telephone
     - Required certificates (certificate type, public key)
     - Description
2. Home Component Directory
   - The Administrator approves the request.
   - The Component Directory issues the requested certificates and store them.
3. Endpoint/Broker
   - Downloads the certificates from its home Component Directory.

> Sends full information about its own configuration (code, contact information, certificates, message paths …) to the home Component Directory.

4. Home Component directory

> Authenticates the component using the stored authentication certificate.

> Stores the complete component configuration.

> From this moment, the ECP network knows the component, which may use all ECP functions.

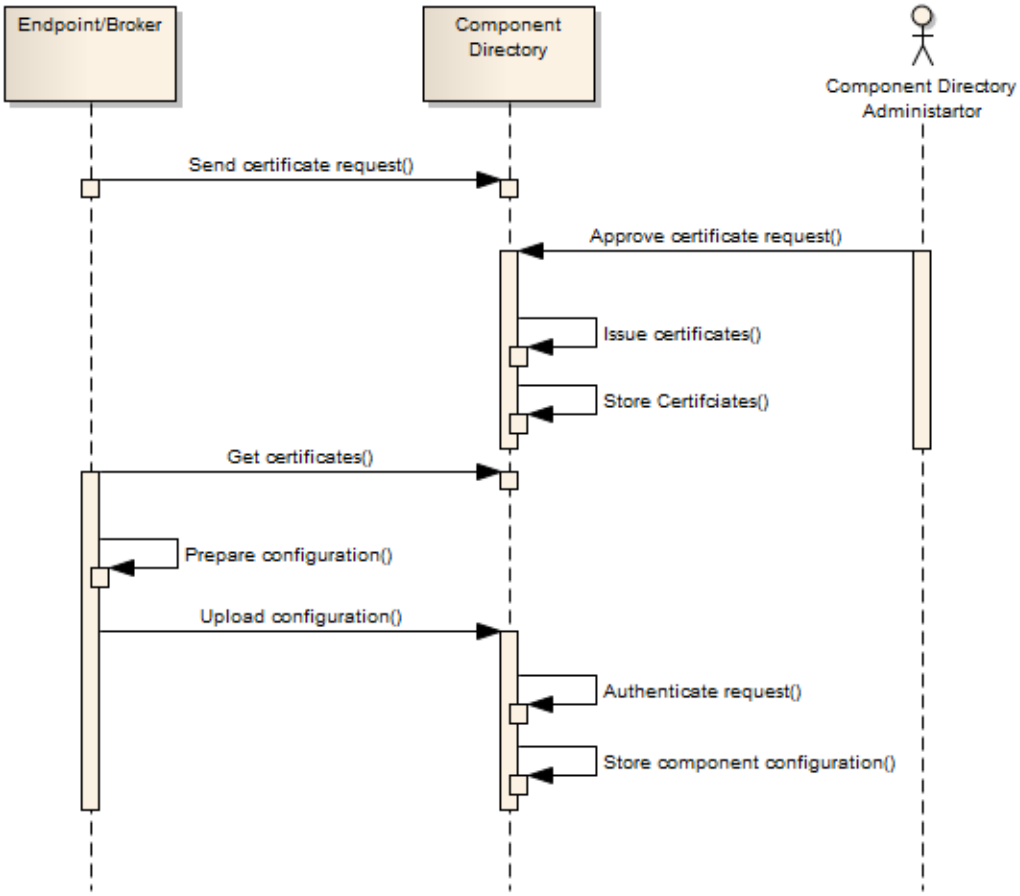


Figure 37: Component Directory UML

4.6.2. Component Registration Management

The administrator of a Component Directory must manually confirm any new registration of an Endpoint or a Broker to the ECP network or any request for certificate renewal from a component without valid certificates. Automatic approval only happens when the request to renew the certificates comes from an already registered component with existing valid certificates.

The Component Directory implements a set of GUI screens to support the registration process.

4.6.2.1. UC Requests to Approve

The Component Directory displays on the "Components page" the list of the registration requests waiting for a manual approval. The Component Directory administrator uses the list to easily view all pending requests and navigate to their detail in order to approve or reject them.

More information on this use case can be found also in section *UC Components*.

4.6.2.2. UC Registration Request Detail

The request detail page shows complete information about a registration request and enables the component directory administrator to perform actions over the request.

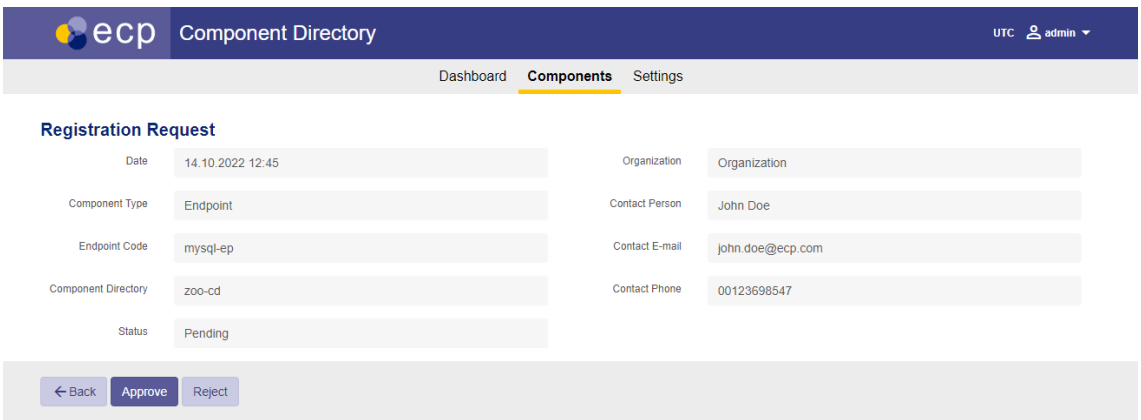


Figure 38: Registration Request GUI

If the request is still in the initial, pending state, the administrator can approve or reject the request using the buttons available on the request detail page. Clicking on the Approve or Reject button shows a modal dialog as these actions require a confirmation by the administrator.

Figure 39: Approve Request GUI

The modal dialog contains a text field where the administrator can optionally enter a reason for his action. This information will be available to the person who created the request. The administrator confirms selected action using the Confirm (or Reject) button on the dialog or cancels the action by using the Back button.

When the request is approved, the Component Directory issues a certificate set for the component and from that moment, the component can push its configuration into the component directory to start using ECP.

If the request is approved and the whole component is revoked anytime later, the request is set into a Revoked state. When a request is in the Revoked state, its detail page contains a Re-approve button that allows the component directory administrator to revert the revocation action (e.g. if the component was revoked by a mistake).

4.6.3. Component Management

The Component Directory administrator is responsible for the management of configuration data of the components for which it acts as the home Component directory (hereinafter referred as its “home components”).

The Component Directory administrator cannot modify data configuration (for example the message paths) of any component (even of its home components).

However, the Component Directory administrator can revoke any certificate used by one of its home components or revoke any such component as a whole. The revoked certificates and components are then no longer included in the synchronization process. A revoked component or a certificate may be reapproved later.

Revoked component can still messages, but they will fail at a receiver’s side. Other components cannot send messages to a revoked component because the revoked component is removed from their components list.

Revoked component does not know about its revocation, because there are no changes in the status of its certificates in the database. In the database of the Component Directory, certificates of the revoked component are marked as revoked. Also, the component has a revoked status in the CD GUI.

Revoking a component means that its certificates are revoked, and that the component data configuration is removed from the component directory. This also means the revoked component cannot upload modified data configuration to its home component directory anymore. In this situation, either the revoked component administrator starts a new registration process (to obtain new authentication certificates), or the Component Directory administrator can reapprove the component.

4.6.4. Component Directory Synchronization

The Component Directory synchronization is used to distribute information about all registered components to Endpoints, Brokers, and other Component Directories. The synchronization interface is using the pull communication pattern. This means that synchronization is always initiated by the component that wants data.

4.6.4.1. Content ID

Since it is expected that the synchronization runs on a cyclic basis it is important to define a way to check if there are any changes in data. To provide this ability each Component Directory holds information about an ID of its used to check if there are any changes. Each Component Directory uses a content version (e.g. an integer number) as content ID. This content version is incremented every time there is a change in the Component Directory content. It is important to notice that the content ID only changes when configuration data of one of its home components changes.

As an example, let us have Component Directory 1 without any registered components (content ID is 0) and Component Directory 2 with already some registered components (content ID is 10).

1. Endpoint 1 registers to Component Directory 1.
2. Component Directory 1 content ID is increased to 1.
3. Component Directory 1 synchronizes with Component Directory 2. Component Directory 1 stores all the components registered with Component Directory 2. Since there are no changes in its home components (currently is only Endpoint 1 home component) the content ID is still 1.
4. Endpoint 2 registers to Component Directory 1.
5. Component Directory 1 content ID is increased to 2.

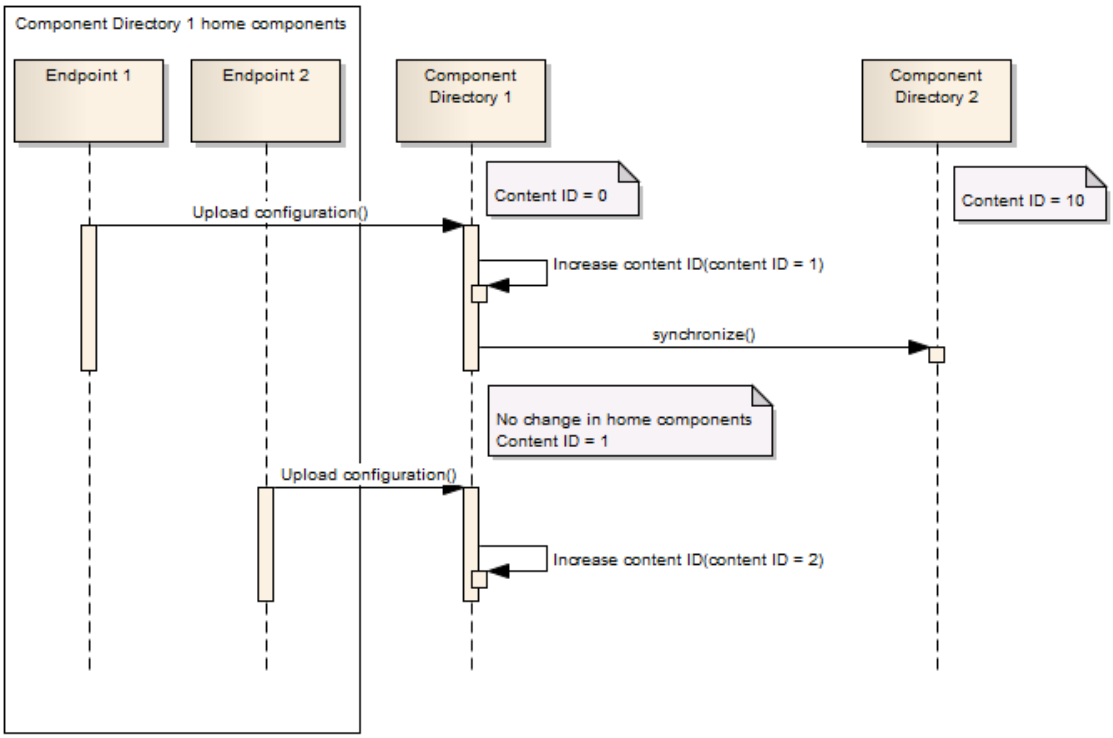


Figure 40: Content ID UML

4.6.4.2. Time to Live

Data configuration of a component (say “C”) locally cached on another component can only be used only for a limited time. This time is known as the “Time to Live” (hereinafter TTL). The TTL is managed by the home Component Directory of the C component and it cannot be changed by any other component. When a Component Directory is requested to provide data about its home components, one part of the response is the TTL for returned data, which is calculated using formula: TTL = now + configured interval. When a Component Directory is requested to provide data coming from another Component Directory, it simply forwards the TTL value received from the last synchronization with that Component Directory.

4.6.4.3. Synchronization

The synchronization is a request/response operation. The synchronization request contains:

- The list of the already cached Component Directories content. This list consists of couples:
  - Component Directory code
  - Content ID
- A flag if all components data or only home components data is requested (this flag is set to the value "only home components data" in the use case of a directory/directory synchronization).

The Component Directory receives this request.

ALL COMPONENTS DATA IS REQUESTED

The Component Directory go through its cached data from other Component directories and its own content and for each:

- Checks by content ID if the requestor has the actual version (content ID is different, or it was not in request).
  - If not, the whole component data from this specific directory is added to the response.
- Add new TTL for the whole component data from this specific directory to the response

If there is in the request a Component directory code that is not known by the Component Directory (for example: Component Directory was removed from Component directory list), the corresponding Component Directory data and TTL are not included in the response.

ONLY HOME COMPONENTS DATA IS REQUESTED

- Check by content ID if the requestor has the actual version (content ID is different).
  - If not, all home components data is added to the response.
- Add new TTL for home components data to the response

4.6.5. UC Setup Component Directory

The Component Directory must first setup itself to be able to communicate in the ECP network. A GUI wizard leading the user through the setup process supports this initialization step.

The setup is a two-step process.

4.6.5.1. Import Certificates

The form on the first step has a single mandatory field requiring the user to select certificate file needed to initialize the Component Directory.

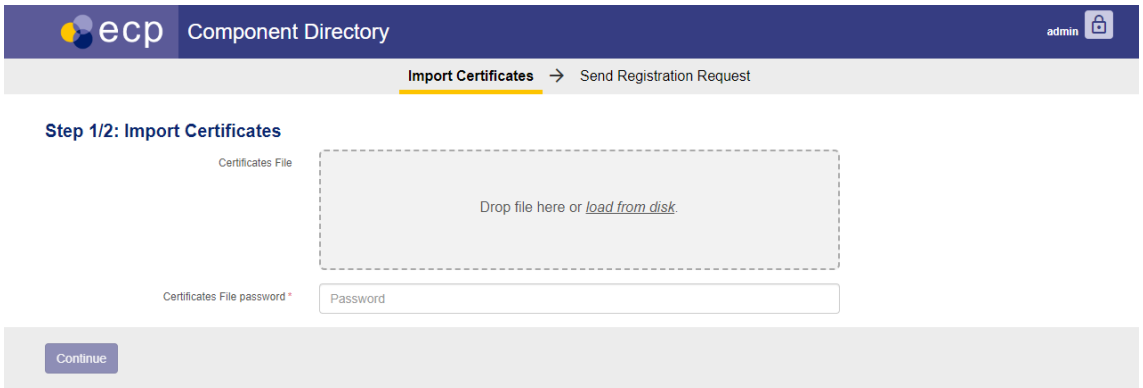


Figure 41: Import Certificates GUI

1. The user inserts the certificate file into the form. The file must contain the certificates of the root certificate authority, the integrated certificate authority including the private key and registration certificate.
2. The user clicks on the "Continue" button.
3. The software verifies the format of the file and if an error occurs, it is displayed directly on the page.
4. The wizard proceeds to the next step.

4.6.5.2. Database Restore

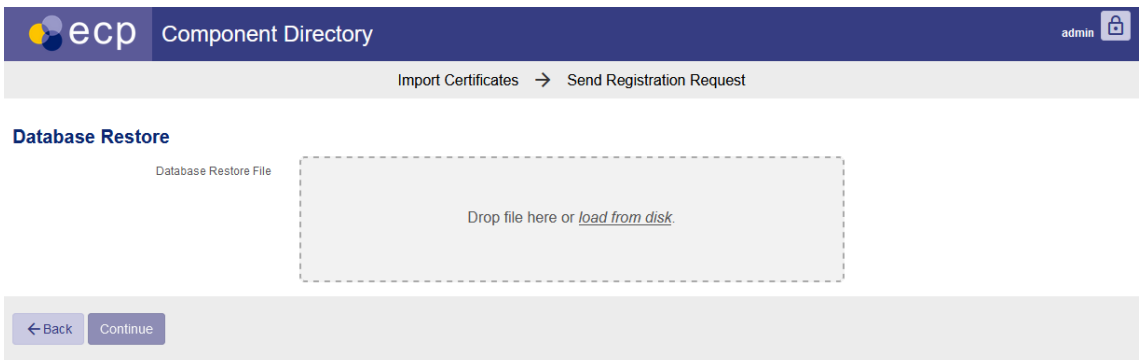


Figure 42: Database Restore

10. Instead of the proceeding with the registration, the user has an option to restore the Component Directory from a previously created database backup.
11. The user clicks on the “Restore Database” button.
12. A new page is opened, with a form to upload a database backup.
13. The user inserts (i.e. drag and drop) a database backup zip file into the form.
14. The software validates the zip file content.
    > The zip file content is described in the *Administration Guide*.

15. The software imports the database backup and replaces a content of the backed-up database tables with a content from the backup.
    > The operation is atomic and if there is an error anytime during the database restoration, no changes are saved, and the error is shown directly on the page.

4.6.5.3. Configure Component Directory

The second step in the setup wizard asks the user for information about the Component Directory and organization.

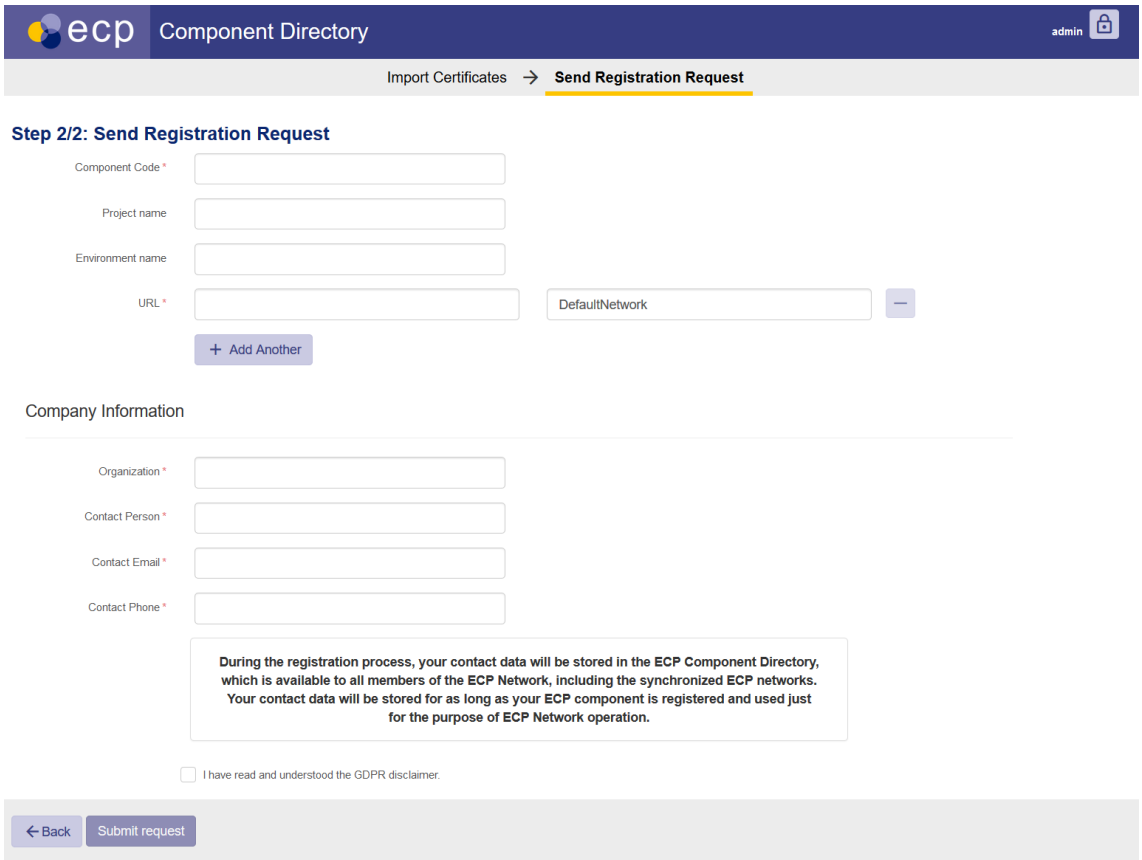


Figure 43: Configure Component Directory GUI

1. The user fills in the form, which contains following mandatory fields:
    > Component code – the code of the Component Directory
    > URL – the list of URLs on which the Component Directory is accessible. At least one URL must be set.
        - Each URL is accompanied with the code of the physical network to which the URL belongs (e.g. Internet). By default, this value is filled to “DefaultNetwork” and it is recommended to change it only when needed. See the management of multiple networks topic in the System Design document for more details.
    > Organization – the name of the organization owning the Component Directory

- Contact person – the name of the person responsible for the Component Directory (Component Directory administrator)
- Contact email – the email of the person responsible for the Component Directory
- Contact phone – the phone number in international format of the person responsible for the Component Directory
- The user clicks on the “Confirm” button.

2. The user can fill in the optional fields:
   - Project name - it will be displayed in the ECP header
   - Environment name - it will be displayed in the ECP header
3. The user clicks on the “Confirm” button.
4. The software validates that all data was entered and meets required format.
   - If the validation fails, the software shows the validation errors to the user and asks the user to correct them before saving the settings.
5. The software stores the information locally and redirects the user to the Dashboard page.

4.6.6. UC Login

The access to the GUI of Component Directory is protected since it provides advanced functionality, which has influence on the ECP network. A user accessing the GUI of the Component Directory must always authenticate using a username and a password to access any page.

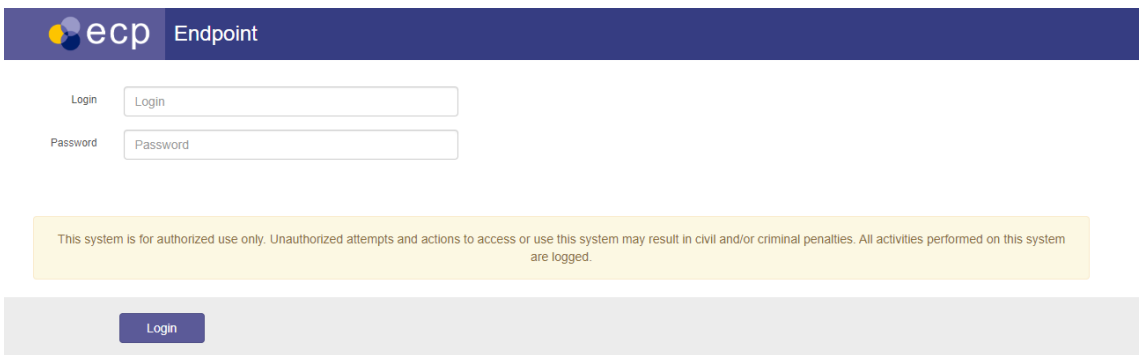


Figure 44: Login GUI

The login page has two fields – username and password. If the authentication is not successful, the software informs the user about the error on the same page. Once the user is authenticated, he can logout through a link in the menu. When the user logs out from the application, he is redirected to the login page.

4.6.7. UC Dashboard

The dashboard is landing page for the Component Directory. It provides basic information and statistics about running Component Directory. It shows following information:

- Basic Component Directory information
  - Component code
  - Description
  - Version of ECP
- Information about the pending certificate requests (registrations or certificate renewal) requiring manual confirmation and a link to the Settings page.
- Information telling if the certificates and the certificate authorities are valid, and when the next certificate expiration happens and a link to the Settings page.
- Information about the number of active components registered within the Component Directory and a link to the Settings page.
- Information about the status of synchronization with other Component Directories (OK - green / attention - red) and a link to the Settings page.
- Information about last runs of background jobs
  - Number of completed jobs
  - Number of jobs waiting for run
  - Number of failed jobs
- Information about the Component Directory readiness. Component Directory is ready when it is registered, authentication certificate is not expired, and it is not reloading a configuration. Readiness tile is not displayed when the readiness endpoint is not enabled in the configuration properties.

4.6.7.1. GUI

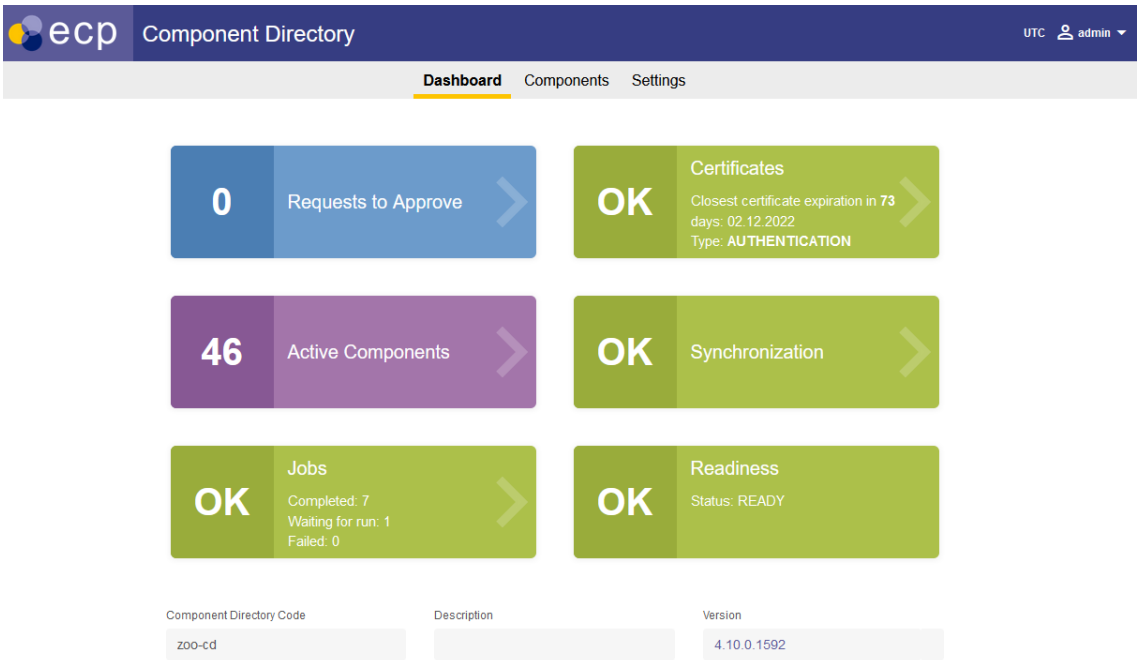


Figure 45: CD Dashboard GUI

4.6.8. UC Components

The Component Directory provides a page with an overview of the components registered within the ECP network or asking for a registration at this Component Directory.

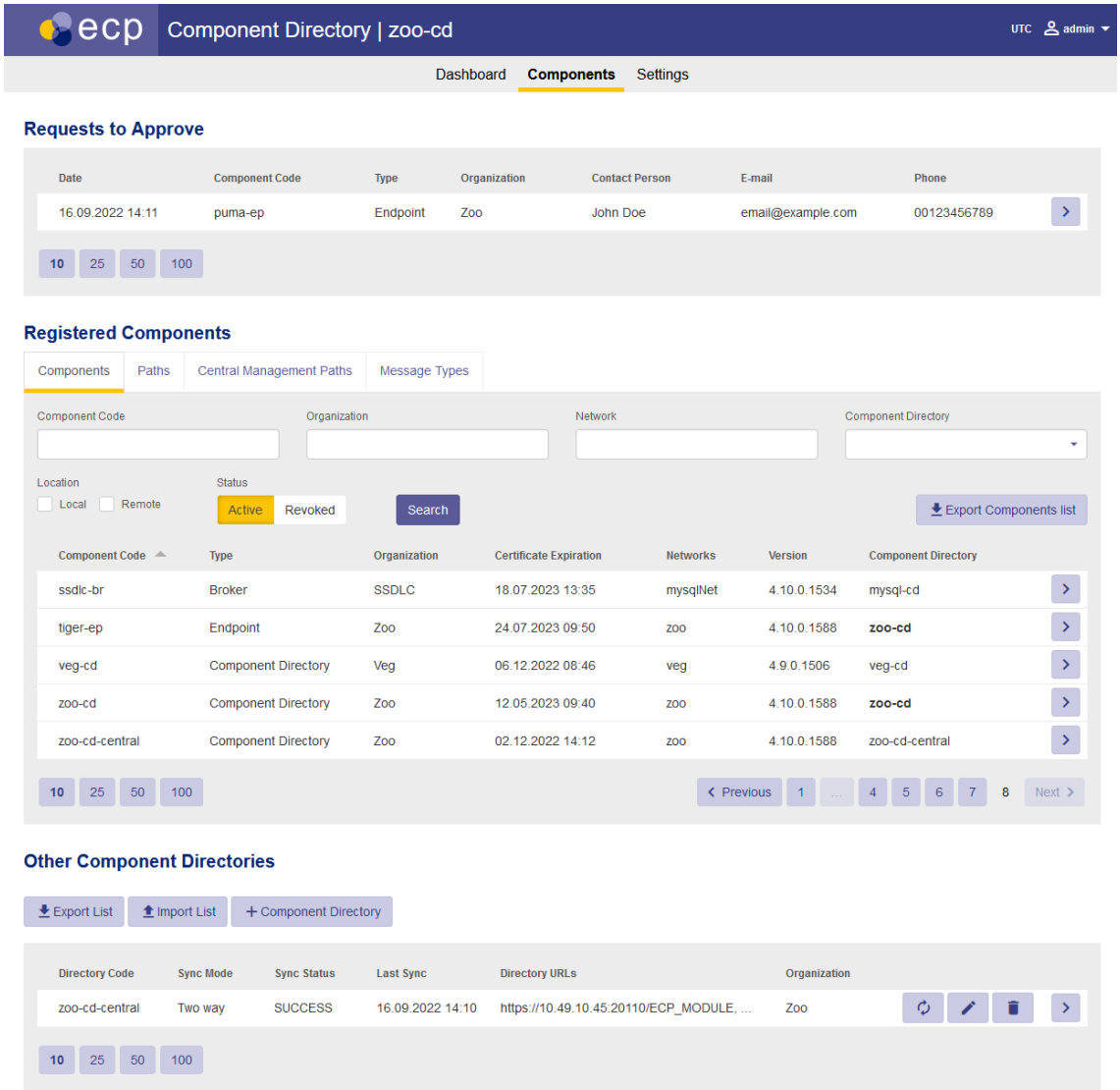


Figure 46: Components GUI

The page contains following information:

- List of requests waiting for approval by the Component Directory administrator
  - > Request date, component code, component type and contact details are shown for each request together with a detail button leading to the request detail page
- List of all Endpoints and Brokers registered in the whole ECP network
  - > Component code, component type, location and contact details are shown for each component together with a detail button leading to the component detail page or a remove button when component has not pushed its configuration yet
  - > Switch button allowing to show the revoked home components – a detail button leads to the detail of the last approved registration request of the revoked component instead of the component detail page

> Export button, which will create a CSV file with information about each component. Each row represents one component and contains values of these columns (column headers are in the first row):
    - Component type
    - EIC code
    - Registered to Component Directory
    - List of networks
    - ECP version
    - Created (time when the component was registered)
    - Last Modification (time when the component pushed its configuration)
    - Organization
    - Last synchronization time (only for the component registered on this CD)
    - Number of sent messages (only for the component registered on this CD)
    - Number of received messages (only for the component registered on this CD)

- List of all message paths in the whole ECP network
  > Receiver, message type, intermediate component and validity interval is shown for each message path together with a detail button leading to the message path detail page
- List of remote component directories
  > Component code, synchronization mode, synchronization status, time of the last synchronization, directory URLs, name of the organization and allowed CDs for synchronization are shown together with a edit button, remove button and detail button leading to the detail page of the component directory and synchronize button to synchronously invoke synchronization with the selected Component Directory
  > Button to synchronize content with all component directories
  > Buttons to export and import the Component Directory list (see Component Directory Registration for more details)
  > Button to add new Component Directory to the synchronization list

4.6.8.1. Component Directory Synchronizer

When a Component Directory is configured to act as a Component Directory Synchronizer (CDS), following changes are made in UC Components:

- "Requests to Approve" and "Registered components" GUI sections from the Components tab are not displayed
- "Export List" and "Import List" buttons from the "Other component directories" section in Components tab are not displayed, changes of configuration are allowed only via GUI
- "Pending edits" and "Pending removals" tables are displayed in the Components tab

When CDS administrator edits or removes CD in “Other component directories”, a new record is created in "Pending edits" or "Pending removals" table, which contains a datetime of the change/removal and admin username. Each row in the pending tables contains "Accept” and "Reject" buttons. "Accept" button is visible only for administrators, which are different than the administrator who made the change/removal. After the change/removal is accepted, a record is removed from the pending table and the corresponding CD record is updated or deleted from the "Other component directories" table.

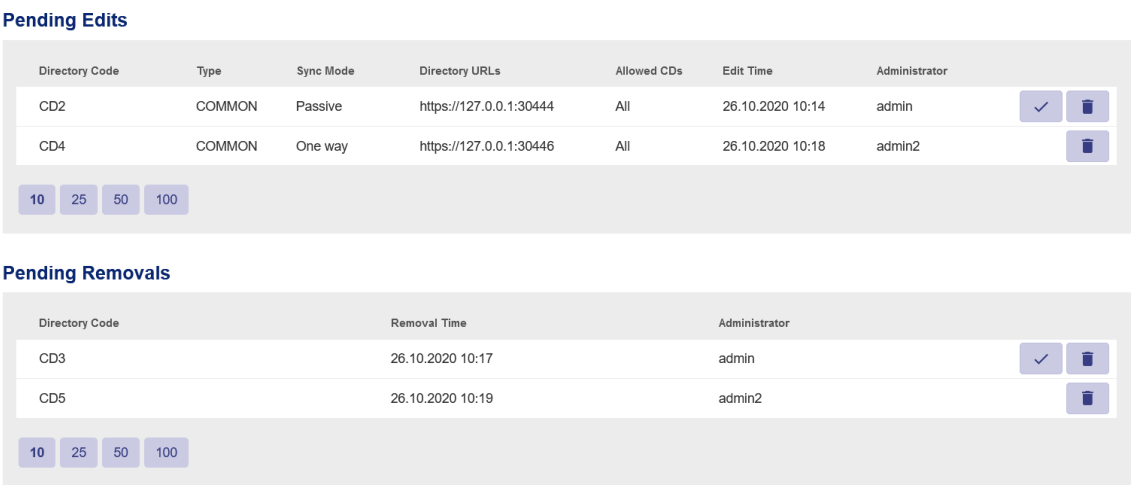


Figure 47: Pending tables in CDS

4.6.8.2. UC Component Detail

The component detail page shows information about a selected Endpoint or Broker and allows the administrator of its home Component Directory to revoke the component.

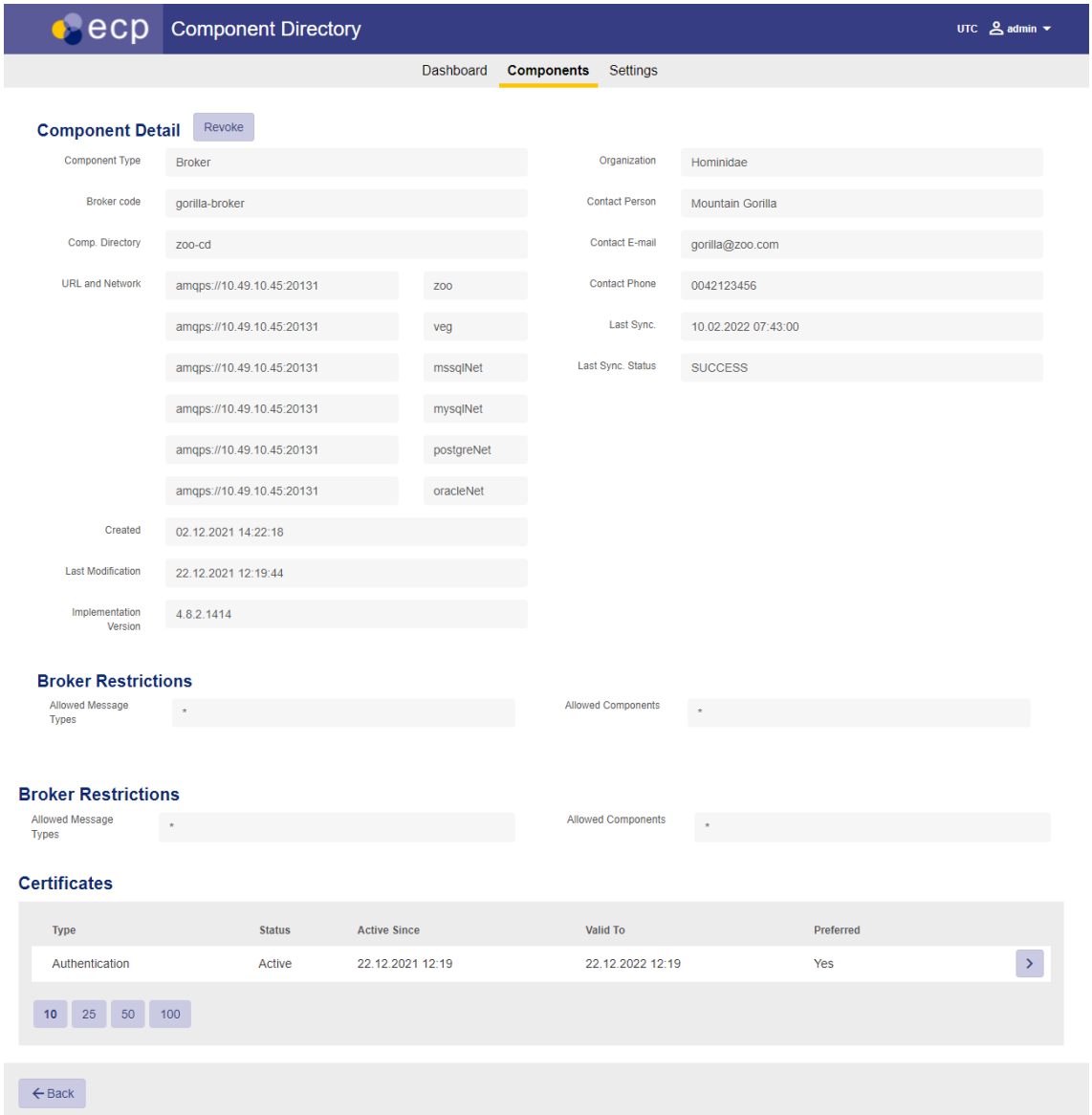


Figure 48: Component Detail GUI

Following information is displayed on the page:

- Information about the component: component code, URLs, type of the component, contact details, home Component Directory, the available physical networks, the date when the component registered and the date when its configuration was last modified. For Broker or Endpoint component the last Component Directory synchronization time and status is shown.
- Button to revoke the whole component. This button is available only in the home Component Directory of the selected component. An administrator cannot revoke a component registered with another Component Directory.
  - > The revocation of a component removes it from the Component Directory, revokes all its certificates and moves its "registration status" to the "revoked" state. Such component cannot communicate within the ECP network until it is approved again. This action requires confirmation through a modal dialog.

- List of message paths defined by a given component. This section is not available for Brokers.
  - > Message type, intermediate component and validity interval is displayed for each message path together with a link to the detail page.
- List of component's certificates.
  - > Type of the certificate, status, certificate authority that signed the certificate, validity interval and information if the certificate is the preferred one (i.e. default) is shown for each record together with a link the detail page.
- Messaging statistics overview for the selected Endpoint in the home Component Directory
  - > List of endpoint messaging statistics including number of sent/received messages and messages waiting to be received / delivered.
  - > List of messaging statistics for each endpoint's connected remote component. Remote components connection status is displayed. Sum of received / sent messages together witch last time of receive / sent operation is shown.

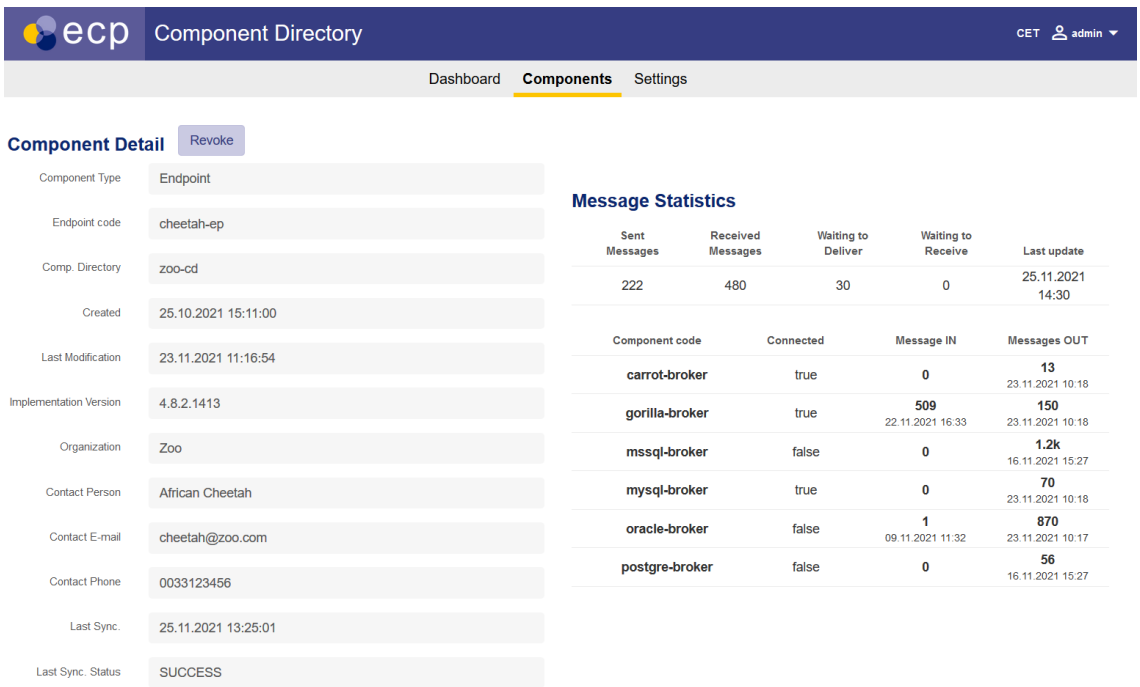


Figure 49: Component Detail GUI 2

4.6.8.3. UC Component Directory Detail

The "Component Directory detail" page shows information about the selected Component Directory and its home components.

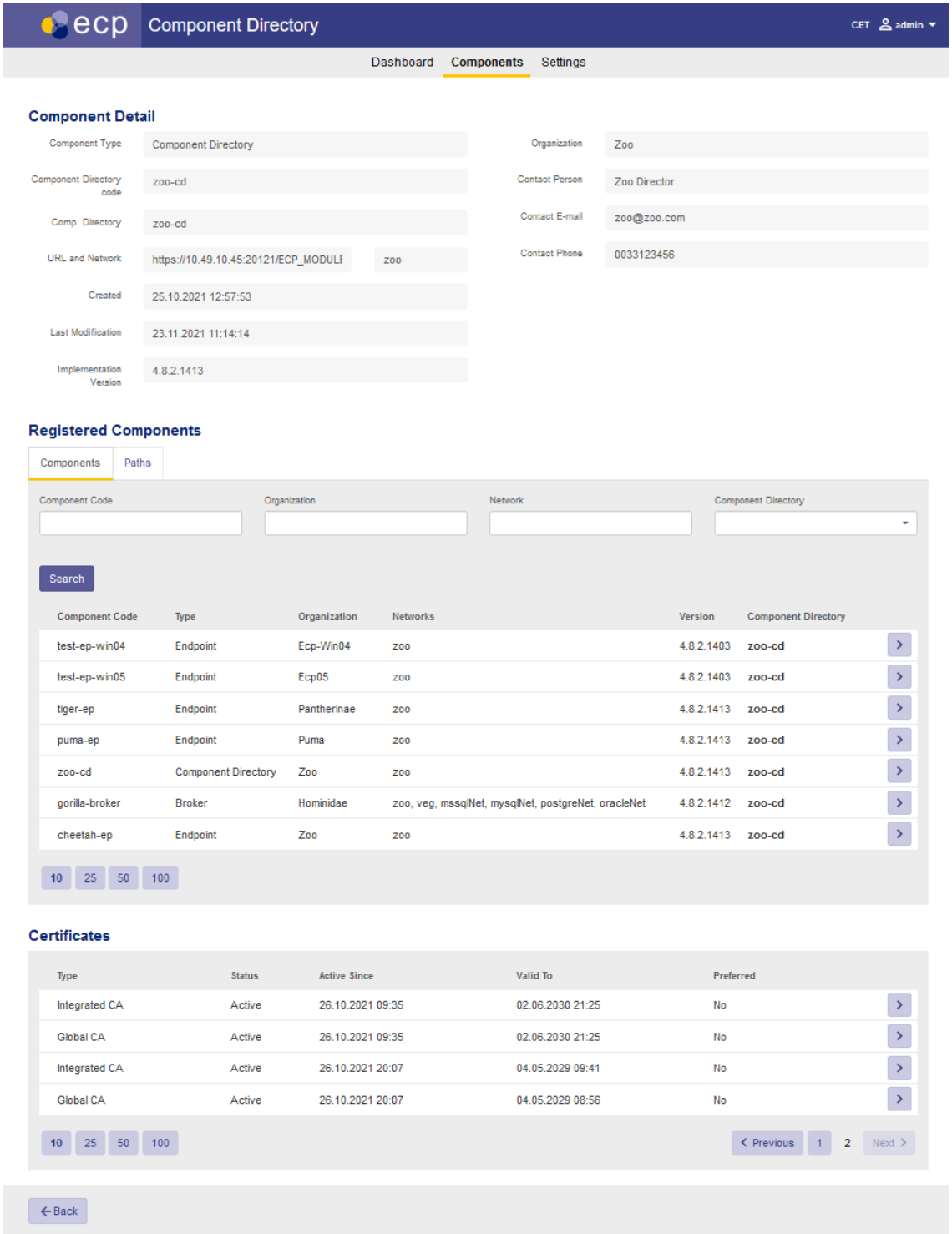


Figure 50: Component Detail 3

Following information is displayed on the page:

- Information about the Component Directory: component code, URLs, available physical networks, and contact details.
- Status of the synchronization with the selected Component Directory.

- Components registered with the selected Component Directory.
  - > Component code, type and contact information is shown for each record together with a link to the detail page.
- Message paths of the components registered with the selected Component Directory.
  - > Receiver, message type, intermediate component and validity interval is shown for each message path together with a link to the detail page.

4.6.8.4. UC Message Path Detail

The "message path detail" page contains all information about a selected message path.

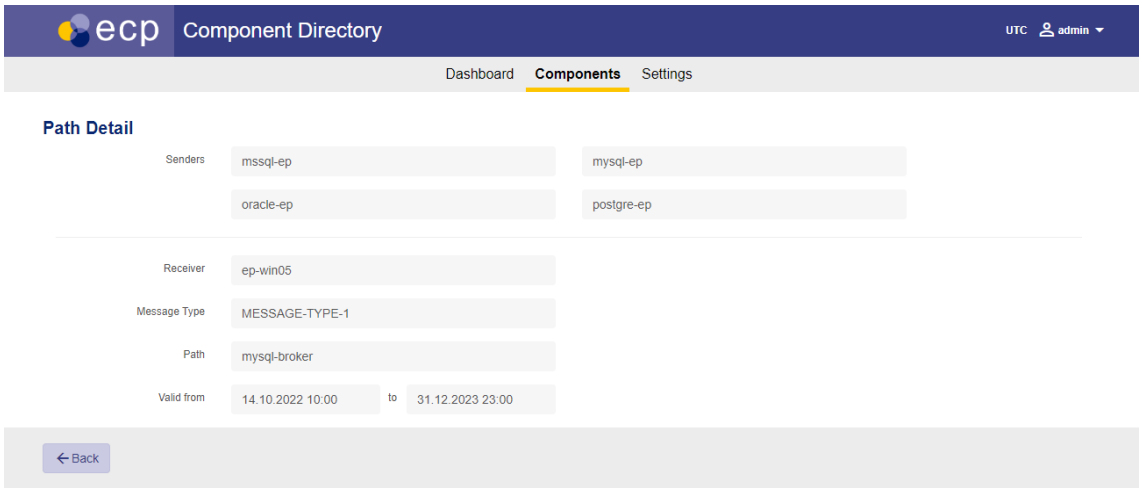


Figure 51: Path Detail GUI

Following information is displayed on the page:

- Validity period
- Allowed senders on the message path
- Receiver, i.e. Endpoint that defined the message path
- Message type routed through this message path
- Optional intermediate component

4.6.9. UC Settings

The "Component Directory settings" page allows the user to manage the certificates and the Component Directory configuration.

4.6.9.1. Basic Scenario

1. The user opens the Settings page in a web browser (via a link in the menu).
2. The Settings page is displayed containing following information:
   - Names – can be switched to editable mode to change the values
     - Project name and Environment name
   - Company information – can be switched to editable mode to change the values
     - Company name and contact information (person, e-mail, and phone)
   - Certificates
     - List of the certificate authorities used by this Component Directory with a link to the detail page
     - List of all authentication certificates of the Component Directory with a link to the detail page
     - Button for importing certificates from a file (typical use is for updating the root certificate authority)
   - Runtime configuration
     - Button to export the current Component Directory configuration to a file
     - Button to import the new Component Directory configuration from a file
     - Button to edit the current Component Directory configuration in GUI
   - Database Backup
     - Button to create a database backup
     - Button to restore from a database backup
   - Time Zone
     - Select box for configuration of time zone
     - Save button for confirming change of time zone
   - Background Jobs
     - Job – background job name
     - Status – job can be in the following states:
       - Completed – the last run was finished successfully
       - Waiting for run – job has not yet started
       - Failed – there was an exception in the last run
       - Disabled – job is not enabled in the configuration properties
     - Start – time when the last run started
     - End – time when the last run finished

- Duration – difference between start and end time
- Failure – error message, only displayed when a job is in the Failed status. Only a beginning of the message is displayed, the whole message is displayed in a modal window after a click on a detail button.

4.6.9.2. Alternative Scenario – Import Certificates

Administrator imports certificate into the local certificate store. This is typically used to insert a new root certificate. The imported certificates are not part of the directory content.

1. The basic scenario is followed up to step 2.
2. The user clicks on the “Import certificates” button.
3. The software opens the file input dialog box and asks the user to select a file for import.
4. The user inserts the certificate file to import and confirms the selection. More information about the keystore file can be found in section *Installation Certificate Set*.
5. The software validates the correct format of the file and its content.
6. The software imports the contained certificates into the local certificate store.

4.6.9.3. Alternative Scenario – Import Configuration

1. The basic scenario is followed up to step 2.
2. The user clicks on the “Import configuration” button.
3. The software opens the file input dialog box and asks the user to select a file for import.
4. The user inserts the configuration file to import and confirms the selection.
5. The software validates the correct format of the file and its content.
6. The software imports the configuration into the database and reloads the application context.

4.6.9.4. Alternative Scenario – Export Configuration

1. The basis scenario is followed up to step 2.
2. The user clicks on the “Export configuration” button.
3. The software opens the file output dialog box and asks the user to select where to save the exported configuration file.
4. The user selects a location for export and confirms.

5. The software exports the current configuration into a file named *"**{COMPONENT-CODE}-configuration.properties**"*.

4.6.9.5. Alternative Scenario – Edit Configuration

1. The basic scenario is followed up to step 2.
2. The user clicks on the "Edit Configuration" button.
3. The software opens a new page with all runtime properties.
4. The user edits the properties and clicks on the Save button.
5. The software validates the correct format of the changed properties.
   > The format and all configuration parameters are described in the *Administration Guide*.
6. The software imports the configuration into the local database and updates the running application.
   > The operation is atomic and if there is an error anytime during the import, no changes are saved, and the error is shown directly on the page. Invalid properties are also highlighted in a red colour.

4.6.9.6. Alternative Scenario – Create Database Backup

1. The basis scenario is followed up to step 2.
2. The user clicks on the "Create Backup" button.
3. The software creates a database backup and exports it into a zip file named *"**{COMPONENT-CODE}_{current timestamp}.zip**"*.

4.6.9.7. Alternative Scenario – Restore Database Backup

1. The basis scenario is followed up to step 2.
2. The user clicks on the "Restore Backup" button.
3. The software displays a warning dialog box.
4. The user clicks on the "Proceed" button.
5. The user selects a database backup zip file to import and confirms the selection.
6. The software validates the zip file content.
   > The zip file content is described in the *Administration Guide*.
7. The software imports the database backup and replaces a content of the backed-up database tables with a content from the backup.
   > The operation is atomic and if there is an error anytime during the database restoration, no changes are saved, and the error is shown directly on the page.

4.6.9.8. Alternative Scenario – Edit Company Information

1. The basis scenario is followed up to step 2.
2. The user clicks on the “Edit” button assigned to the section with company information.
3. The software switches the company information to editable mode allowing the user to change the values.
4. The user makes required changes and confirms them by clicking on the “Save” button.
   > Alternatively, the user may discard the changes using the “Cancel” button.
5. The software saves the changes to the Component Directory.

4.6.9.9. GUI

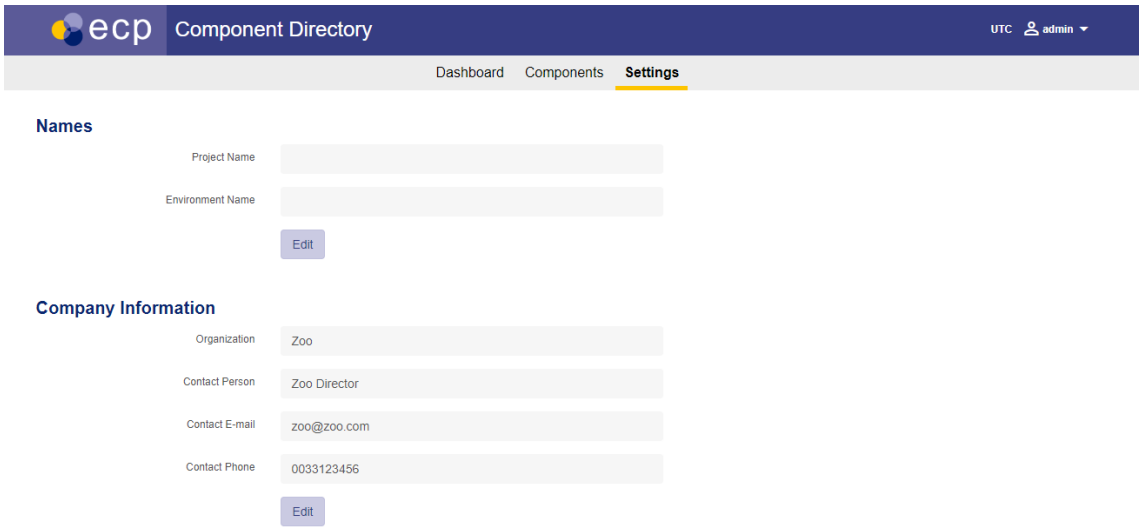


Figure 52: Component Directory - Settings 1

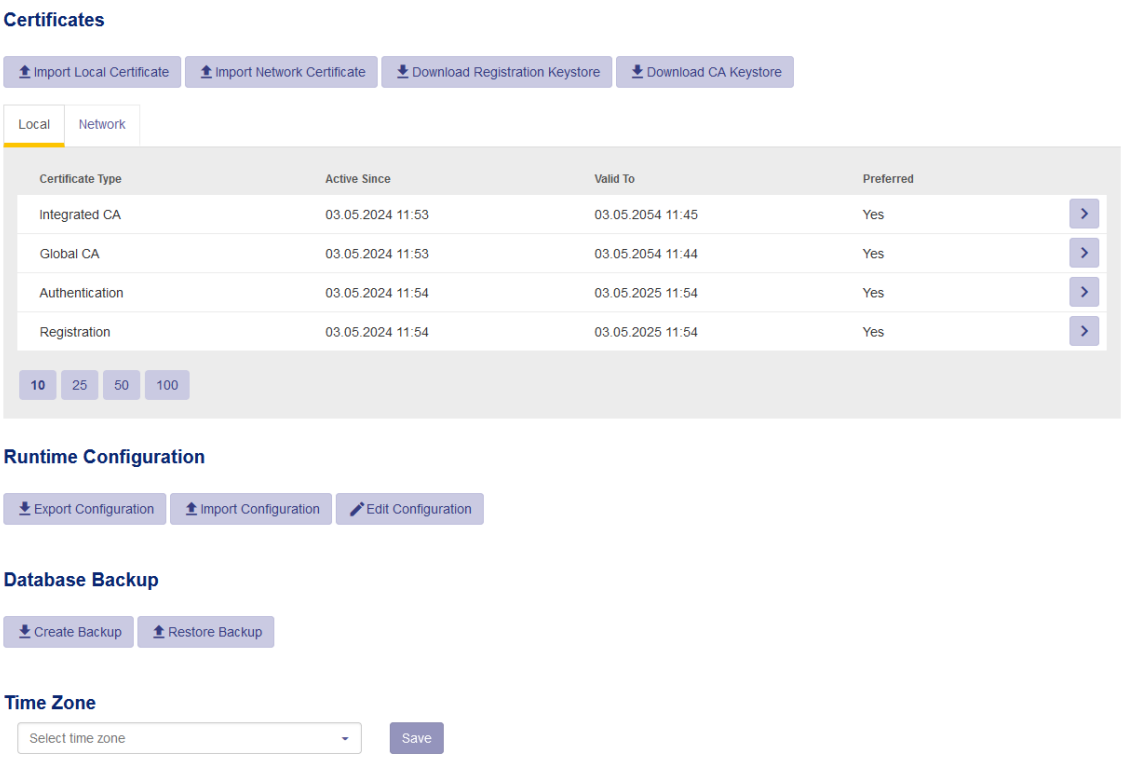
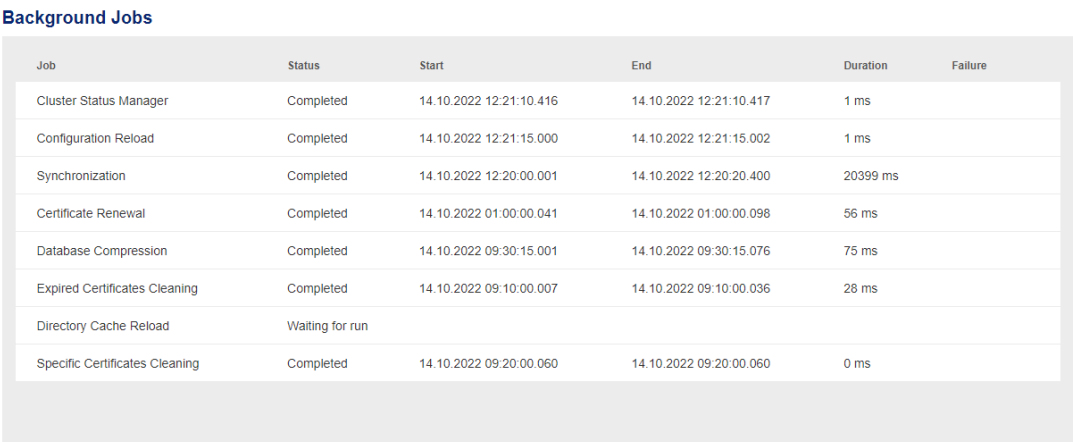


Figure 53: Component Directory - Settings 2



Background Jobs

Job	Status	Start	End	Duration	Failure
Cluster Status Manager	Completed	14.10.2022 12:21:10.416	14.10.2022 12:21:10.417	1 ms	
Configuration Reload	Completed	14.10.2022 12:21:15.000	14.10.2022 12:21:15.002	1 ms	
Synchronization	Completed	14.10.2022 12:20:00.001	14.10.2022 12:20:20.400	20399 ms	
Certificate Renewal	Completed	14.10.2022 01:00:00.041	14.10.2022 01:00:00.098	56 ms	
Database Compression	Completed	14.10.2022 09:30:15.001	14.10.2022 09:30:15.076	75 ms	
Expired Certificates Cleaning	Completed	14.10.2022 09:10:00.007	14.10.2022 09:10:00.036	28 ms	
Directory Cache Reload	Waiting for run				
Specific Certificates Cleaning	Completed	14.10.2022 09:20:00.060	14.10.2022 09:20:00.060	0 ms	

Figure 54: Component Directory - Settings 3

4.6.10. UC Certificate Detail

The "certificate detail" page shows information about the selected certificate and may provide buttons for changing the certificate state.

Following information are displayed for the certificate:

- The validity period of the certificate
- The type of the certificate, i.e. its usage within ECP (authentication, signing, encryption)
- The current status of the certificate (active, expired, revoked)
- The name of the certificate authority that signed the certificate
- Information if the certificate is currently preferred (default) for the given certificate type

The actions available depend on the certificate status:

- Revoke button if the certificate is in active state
  - > Revocation of a certificate marks the certificate as unusable in the ECP network and therefore the certificate is not included in the synchronization.
- Re-approve button if the certificate is in the revoked state
  - > Re-approval of a certificate moves a previously revoked certificate back to the active state allowing it to be used within the ECP network.

4.6.10.1. GUI

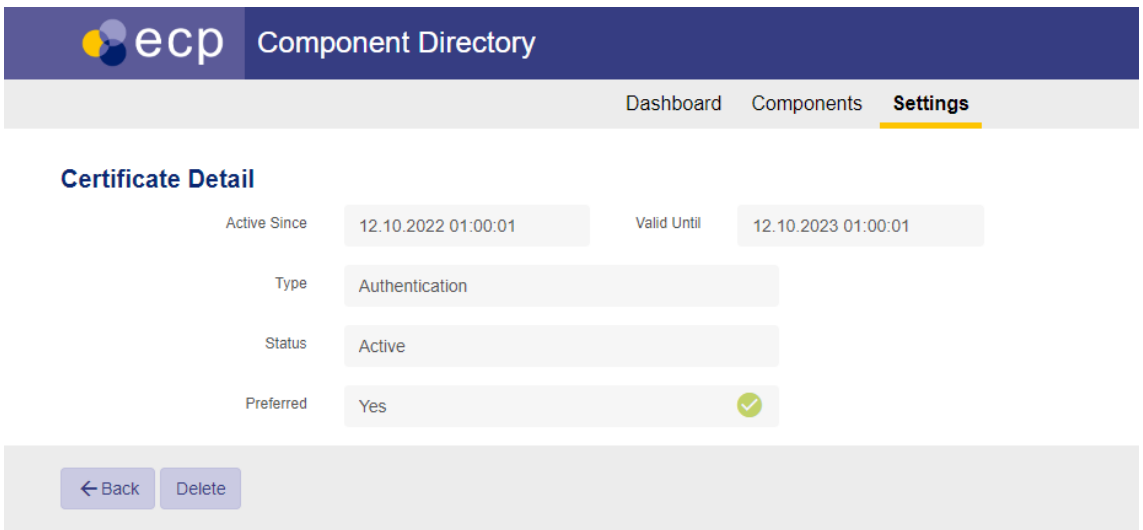


Figure 55: Certificate Detail

4.6.11. Component Directory Registration

When there is a need to include two Component Directories into the same ECP network or to add a new Component Directory in an existing network, the administrators of each Component Directory must perform a manual initialization procedure, which is importing a Component Directory List file provided by the ECP network administrator.

Two user roles participate in the Component Directory setup:

- Component Directory Administrator – This role is responsible for one or more Component Directories. He is responsible for updating information in his own Component Directory based on information provided by the ECP Network Administrator.
- ECP Network Administrator – This role is responsible for the whole ECP network and for managing and distributing the information about the authorised Component Directories to the Component Directory Administrators. Usually, the ECP Network administrator will also be the administrator of one of the Component Directories. It is his responsibility to make sure the list contains correct information about all Component Directories in the network.

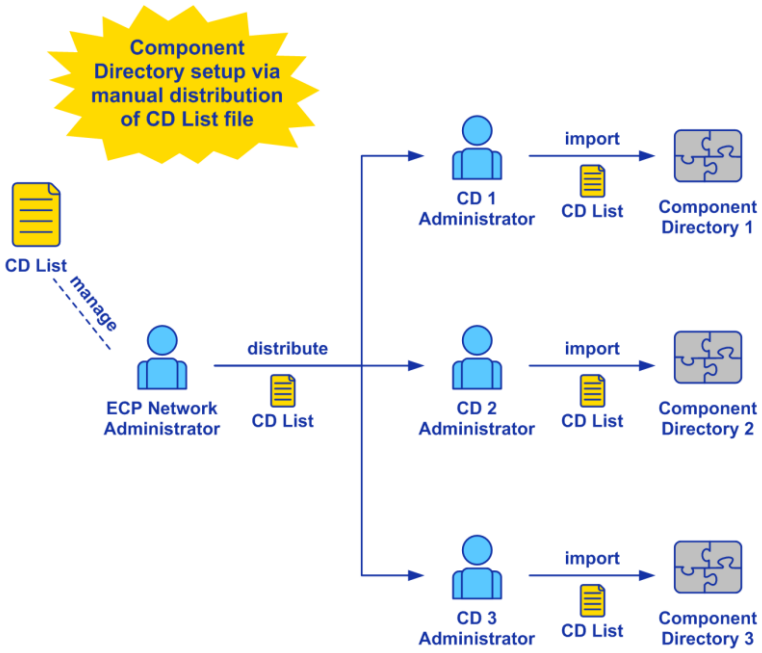


Figure 56: Component Directory Registration

1. ECP Network Administrator initiates the setup / update of ECP network by manually updating the CD List file and distributing it to every Component Directory administrator in the network. The distribution way is not defined by ECP.
2. Administrators of the Component Directories import the provided list into their own Component Directories.
3. The setup of interconnected ECP network is finished when all the Component Directories have imported the CD List. All the Component Directories have the same view of the ECP network and can automatically synchronize their content.

4.6.11.1. Import / Export

The import and export of Component Directory List follows several rules:

- Import
  - > If the list contains information about own Component Directory, that information is ignored during import.
  - > System validates correctness of the provided information and if the validation fails, it rejects complete file without making any changes.
    - – It validates the format and completeness of the file.
    - – It checks there are no duplicities in the codes of Component Directories.
  - > If the list does not have correct format, contains invalid data or contains duplicate information (e.g. two directories with the same code), the whole list is rejected, and no changes are made. ECP informs the administrator about detected errors on the GUI.
  - > Own Component Directory cannot be removed during the import of a Component Directory list even if it is not present in given list.
    - – Other Component Directories are removed during the import if they are missing in the uploaded list together with all their home components.
- Export
  - > The whole view of the ECP network is always exported, i.e. information about own directory and all known remote directories.
  - > Name of the exported file follows pattern cd-list-${ComponentDirectoryCode}.txt, where ${ComponentDirectoryCode} is the code of the Component Directory where the file was exported.

4.6.11.2. Component Directory List File Format

Information about all Component Directories inside the ECP network is stored in a single file and the ECP Network Administrator cares about validity of its content. The file is in a plain text format and contains the component code, primary URL and secondary URL of each Component Directory.

Detailed description of the Component Directory list file attributes is in the table below:

Name	Description
Component Directory code	Code of the Component Directory.
Primary URL	Primary URL of the Component Directory in format according to RFC 1738.
Secondary URL	Secondary URL of the Component Directory in format according to RFC 1738.

Format of the Component Directory list file:

- File must be in UTF-8 encoding.
- Each Component Directory record is delimited by the new line character (LF U+000A).
- Each attribute in the Component Directory record is delimited by an empty space character (SPACE U+0020) for preceding and following attribute.
- Attributes must be stored in the Component Directory record in the following order:
  - Component Directory component code
  - Primary URL
  - Secondary URL (if available)

Find sample Component Directory List in *[Internal API]/componentDirectory/examples/componentDirectoryList.txt*.

4.7. Message Archive

The ECP Endpoint provides the possibility to archive all transferred messages together with basic metadata. This functionality is provided through archive handlers, which can be developed separately using a delivered API and then plugged into the Endpoint. The ECP software comes with a default implementation, a file system archive handler that writes the messages as files into a dedicated folder on the disk file system.

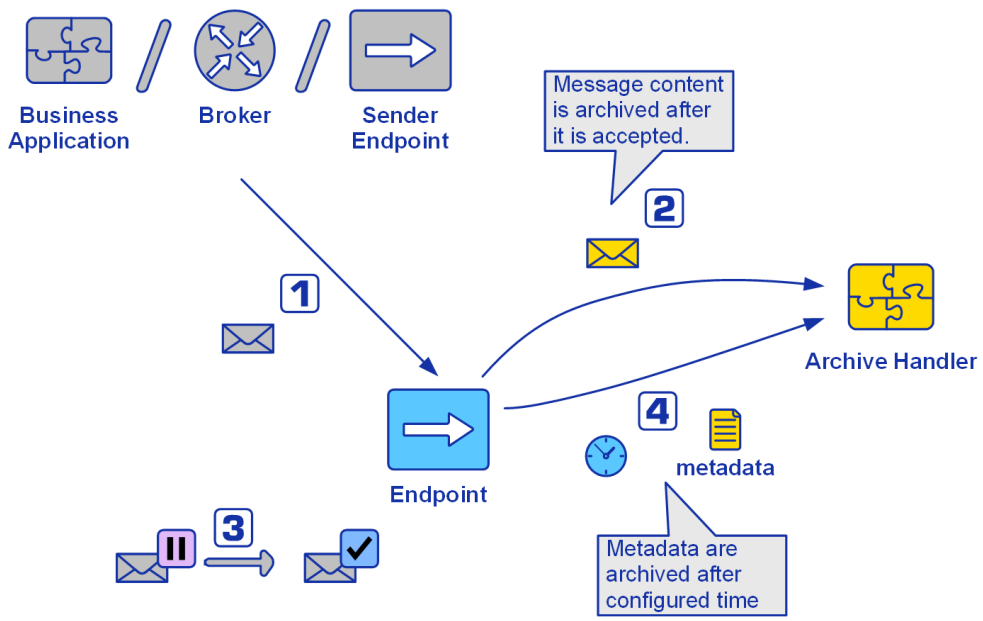


Figure 57: Message Archive

The archive is a two-phase process with the following steps.

1. A message is accepted by an Endpoint from any source (business application, Broker, Endpoint).
2. The Endpoint passes the message content to the configured archive plugin.
3. The message reaches the final state, i.e. it is received by a business application or the delivery fails.
4. The Endpoint archives metadata of the message including acknowledgments and signature.

Information related to the archived message passed to archive plugin are:

- Message content (payload)
- Acknowledgments related to the message
- Basic message metadata i.e. sender, receiver, message type, sending and delivery dates, final state of the message
- Message signature (to be able to proof the origin of the message)

By default, the message archive is disabled, and the administrator should configure the archive functionality, if desirable. When the archive is disabled, the messages are not archived externally and remain in the ECP internal storage for a limited period based on configuration. After this period, the message is deleted completely from the ECP internal storage.

4.7.1. File System Archive Handler

ECP comes with the default implementation of archive handler writing the messages into a configured folder on the file system. It creates two files for each message:

- Message content file – a binary file with the content of business message
- Message metadata file – an XML file containing basic metadata, signature and acknowledgments

The configuration and use of the archive handler are described in detail in *Administration Guide*.

4.7.2. Message Deletion

ECP implements a message deletion functionality to free its internal store from the old processed messages. The message retention period is configurable and can be adapted to the traffic on specific ECP installation. The ECP instances with high traffic will have this period set to lower interval (e.g. several hours) than instances with lower traffic (e.g. several days). By default, the retention period is configured to two weeks.

A mechanism to prevent contentions and slow behaviour of ECP is provided by ECP. If the number of messages stored in ECP internal storage exceeds a threshold, ECP will emit a warning message advising to reduce the message retention period. When the number of messages exceeds a critical threshold, the messaging functions of ECP will be paused until messages amount drops. In such case, user will be warned and will be advised to immediately execute deleting process. Both thresholds are configurable, but only to make it possible to adjust the limits in case of emergencies.

4.8. Version Management

4.8.1. Overview

An ECP Network may contain a large amount of deployed ECP / MADES components. Due to this wide deployment, it is required to allow those components to communicate even in case of different versions. One of the main aims of the version management is to recognize that there is a new version available and to inform the messaging components administrators. From ECP version 3.0, there is also a support for MADES specification versioning.

4.8.2. Versioning Schema

ECP uses a common and widely known version schema based including a major version, a minor version, and a revision number. The schema is: <major version>.<minor version>.<revision>

- Major version is increased when there is significant jump in functionality.
- Minor version is increased when there are only small changes or some significant fixes.
- Revision is increased when there are only bug fixes.

4.8.3. MADES specification versioning support

Versioning support was introduced in ECP version 3 to comply with the MADES specification. Thanks to the well-defined version management, smooth rollout of new versions of MADES specifications is possible. All the Component Directories and Endpoints in network implementing MADES specification of version N will be backward compatible with version N-1.

If the difference between any two communicating components (sender and receiver of a message, Endpoint and Component Directory, Endpoint and Broker) is too big that the compatibility is not guaranteed, the communication is rejected. In such case, the message is moved to the failed state with descriptive error message.

MADES version is included in every message sent from ECP version 3 and higher.

4.9. Other Details

4.9.1. Generating and Processing Acknowledgments

An acknowledgment is special type of message, which is generated as a reaction to some important event related to a business message. Each acknowledgment is related to a business message (further called as its original message) and is sent back to the sender Endpoint of the original message. ECP always routes an acknowledgment using the same message path as its original message and therefore the same communication (direct / indirect). For this reason, a sender Endpoint can send only a business message using the direct communication pattern if it itself allows incoming direct connections.

The following acknowledgment types are used:

- delivery acknowledgment
- receiving acknowledgment
- failure acknowledgment
- tracing acknowledgment

4.9.1.1. Generating and Processing Delivery Acknowledgments

An Endpoint generates a delivery acknowledgement immediately when it accepts a business message.

Each Endpoint that processes a delivery acknowledgment changes the state of the original message in the database to "Delivered".

4.9.1.2. Generating and Processing the Receive Acknowledgments

An Endpoint generates a receive acknowledgement immediately after it successful delivers the business message to a business application.

Each Endpoint that processes a receive acknowledgment changes the state of the original message in the database to "Received".

4.9.1.3. Generating and Processing the Failure Acknowledgments

A failure acknowledgment is generated if a failure occurs during the processing of a business message on an Endpoint. A Broker generates no acknowledgement.

Each Endpoint that processes a failure acknowledgment changes the state of the original message in the database to "Failed".

4.9.1.4. Generating and Processing Tracing Acknowledgments

A recipient Endpoint that accepts a tracing message immediately sends a tracing acknowledgment back to the sender Endpoint.

Each Endpoint that processes a tracing acknowledgment changes the state of the original message in the database to "Delivered".

4.9.2. Large Message Transport

ECP does not support the transport of large messages by the means of splitting them into multiple smaller messages on the sender side and reconstructing the original message from the parts on the receiver side. Any content sent over ECP is always sent as one message and the content size must not be bigger than the limit of ECP (50 MB).

4.9.3. Message Priority

The message priority allows the ECP Endpoint administrator to influence the order of the messages during their processing at his ECP Endpoint. High priority messages are likely to be processed and sent through the message path before the low priority ones.

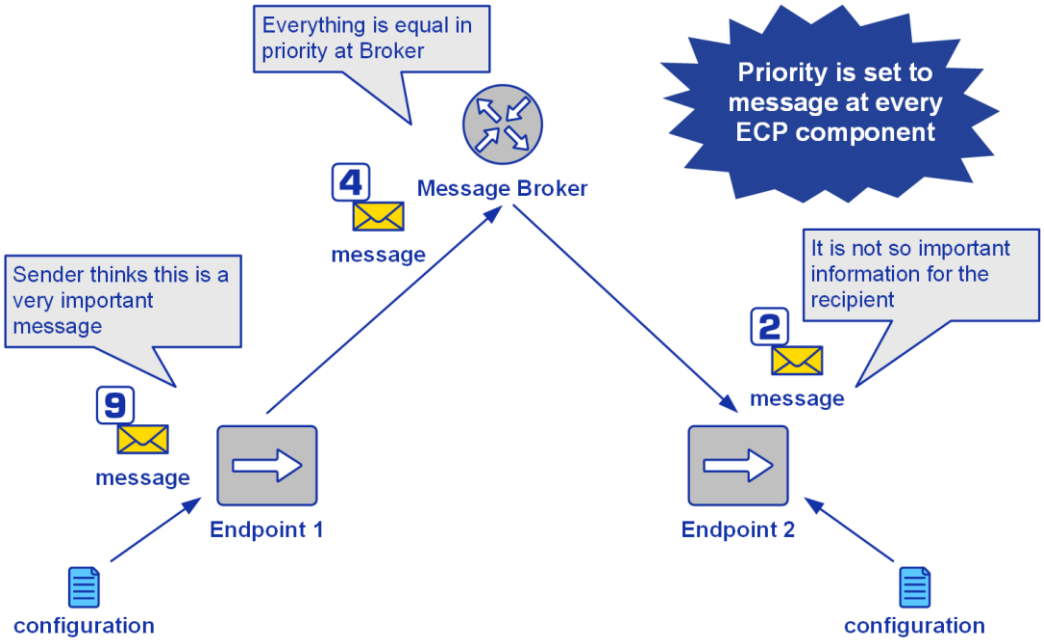


Figure 58: Message Priority

- A priority value is assigned to every message when the sender Endpoint accepts it. The value is set according to the priority configuration value associated to its message type.
- The priority on the Broker is always the same, default priority.
- The recipient Endpoint also assigns a priority to the message according to its own configuration.

ECP supports 10 numeric values for the message priority in range from 0 to 9 inclusive. Higher number means higher priority so priority 0 is the lowest and 9 the highest possible priority. If no explicit priority is configured for a message, ECP uses priority 4 as the default.

4.9.4. Message Expiration

The message expiration is a mechanism to inform the sender that the message cannot be delivered or received because it takes too long time. The expiration period is configurable on the sender side.

The sender Endpoint sets expiration time on each message and according to the configuration when it accepts the message. The messages are expired using the "Message Expiration" component in combination with the AMQP message expiration mechanism. For more details, see System design document.

The expiration results into setting the expired message into the failed state with the appropriate description.

4.9.5. Message Processing Order

ECP does not give any guarantees that messages will be processed and delivered in the same order as they were sent. Therefore, business applications must not rely on the delivery order of messages.

Although the order of messages is not guaranteed, ECP processes messages in the order given by:

- the message priority on each ECP component
- the acceptance time at each ECP component

4.9.6. UC Configuration

An administrator configures an ECP component through an external properties file. This file allows the administrator to configure any of the ECP parameters. Configuration changes using the configuration file requires a manual restart of the ECP component and changes must be made separately and independently on every ECP instance in a failover / cluster deployment.

In addition to the file-based configuration, the ECP Endpoint and the Component Directory allow the administrator to configure some selected parameters through GUI screen (see UC Settings). In this case, the changes are applied without a manual restart of the ECP component.

The screen provides the functionality to export and import the configuration from/to a properties file. The administrator uses the export button to download the current configuration, to review it and to make the required changes. Then he uses the import button to upload the changes. The software validates the correctness of the values in the imported file and applies the configuration.

In case the same parameter is configured both directly in the properties file and via GUI, GUI configuration takes precedence.

The list of the configuration parameters is included in the *Administration Guide* together with their description, default values and allowed configuration source.

4.9.7. Removing a Broker from the ECP Network

When an Endpoint chooses a message path through which it will route a message, it puts the message to its internal queue, as described in section *Endpoint – Send Message*, and this choice of the message path is definitive. If a Broker shuts down and there are messages planned to be transferred through this Broker, Endpoint will try to send them to the removed Broker until they expire. There is no additional attempt to re-route such messages using different message path.

Without message re-routing, it is the responsibility of the ECP network administrator to coordinate necessary actions when a Broker leaves the ECP network to prevent situations when it is impossible to deliver messages due to broken message paths.

It is possible to avoid this situation by using recommended approach.

1. Edit all message paths using the Broker and choose another Broker or switch to direct transport pattern.
2. Leave the Broker running for a while (e.g. a day) after there is no more path through the Broker to allow the transfer of all the messages.
3. Shut down the Broker.
4. If the Broker leaves ECP network permanently, remove the component from the network.

4.9.8. ECP Instance Colour Label

ECP allowed the administrator to visually label his ECP instance with a selected colour. An Endpoint or a Component Directory administrator can select a colour that will be used in the GUI header to label his ECP instance. This label can indicate the environment in which the ECP instance is deployed (e.g. blue for test environment and default for a production).

The administrator configures the label colour in the ECP settings and selects from several predefined values. Details of the configuration is described in the *Administration Guide*.

4.9.9. Installation Certificate Set

ECP is a communication platform focused on security, and as such, it uses X.509 certificates to implement the security features. The X.509 certificates used by the platform must be issued by a certificate authority, which the platform trusts. The trusted certificate authorities can be different for each ECP network.

To initialize the platform with these trusted certificates, the administrator must import the certificate set into its ECP component. This step must be done during the installation of the ECP component and update is later possible through the Settings page.

4.9.9.1. Content of the Certificate Set

The certificate set is distributed in an archive, which is in the JKS format (Java KeyStore). The archive contains certificates; each has its own alias that is used to assess the type of the certificate. The content of the file is described in the table below.

Certificate Type	Alias	Description	Private Key Included

Root certificate	global	The root certificate that is used to sign integrated certificates.	No
Integrated certificate	integrated	Certificate assigned to a Component Directory that is used to sign certificates for the home components of given Component Directory.	No
Registration certificate	registration	Certificate used for authentication and SSL communication during the registration phase of Endpoint and Broker.	Yes

Note that when a certificate is updated, the file should contain only the updated certificate and not the others.

4.9.9.2. Creation of the Certificate Set

The administrator of the root certificate authority for each Component Directory creates the certificate set separately as it contains only one integrated certificate belonging to one Component Directory. The administrator creates the certificate set when he

- issues an integrated certificate authority for a new Component Directory
- renews the root or integrated certificate

The administrator uses a command line tool called “keytool” that is part of the Java Development Kit (JDK) to build the JKS file from standard certificates in DER format. The keytool can import only certificate or a certificate together with corresponding private key.

4.9.9.3. Storage for the Imported Certificates

When importing the certificates from the JKS file, system stores them into its local storage and does not propagate them to the component directory content. The imported certificates can be used only by the component that imported them.

4.10. Message Broker

The "Message Broker" (or simply a "Broker") is a router component, used by the Endpoints, that acts as an optional intermediate component in the message route path. It ensures a reliable and secure communication with the Endpoints using SSL protocol and it authenticates the connecting Endpoints using their authentication certificate.

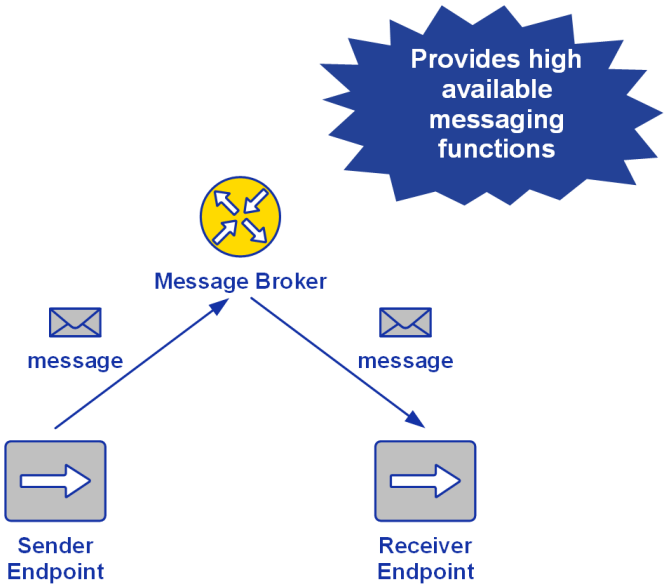


Figure 59: Message Broker

4.10.1. Restricting Message Paths

Every Broker can define a pattern for message paths that can be routed through that Broker. This is useful in preventing the misuse of a business-critical Broker by non-critical, possibly large, messages, which can then slow down the critical process.

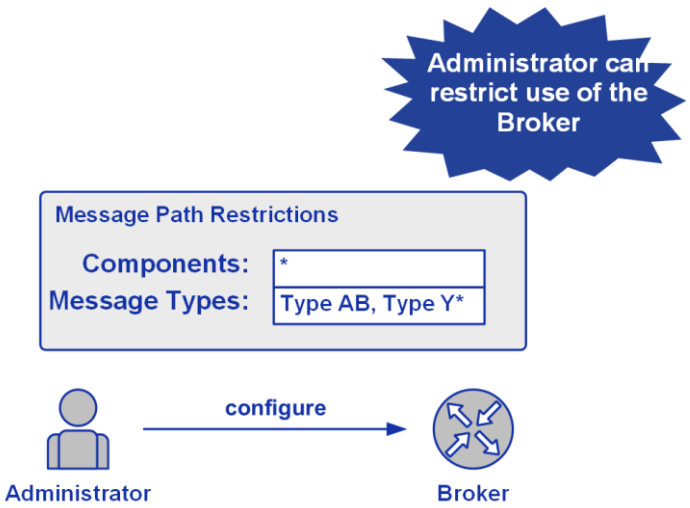


Figure 60: Broker Restrictions

The message path restrictions can be put on messaging components, message types or both. If the field contains a wildcard (*), all values are allowed.

- Components – the list of messaging components allowed using this Broker to send / receive messages.

- Message types – the list of the message types allowed passing through this Broker. Wildcards can be used at the end of the message types.

The management of the message paths restrictions is available to the Broker administrator in the form of file-based configuration that should be changed manually, followed by a restart of the Broker.

4.10.2. Endpoint Authentication & Authorization

The Broker automatically authenticates every connecting Endpoint and authorizes it to access to the queues according to the rules defined in this section.

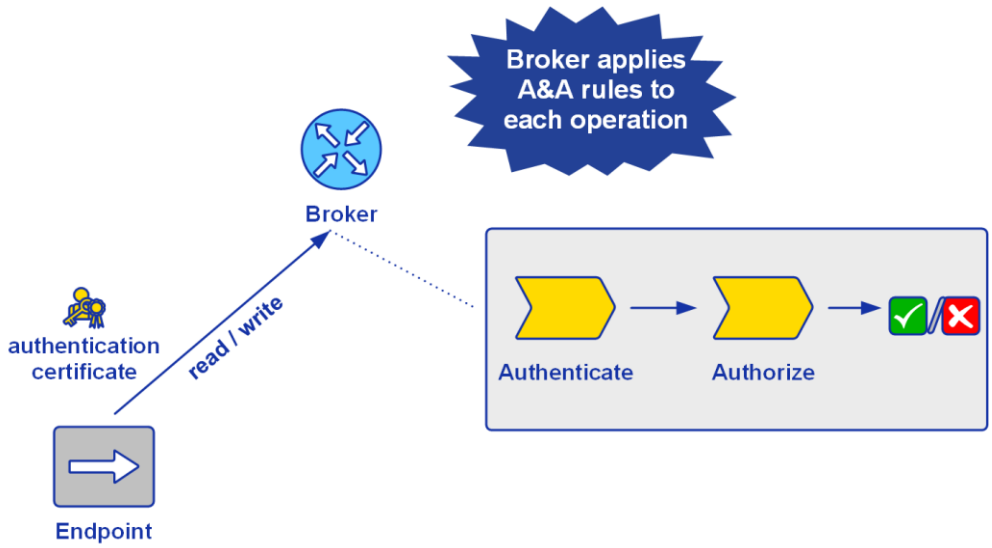


Figure 61: Broker Authentication

4.10.2.1. Endpoint Authentication

Any Endpoint must be successfully authenticated before accessing the Broker. The Endpoint uses its authentication certificate in SSL communication and the Broker uses the provided SSL certificate for authentication of the connecting client.

1. The Endpoint requests a connection to the Broker in order to send / consume messages.
2. The Broker authenticates the Endpoint using its authentication certificate.
3. If the authentication succeeds, the Endpoint is successfully authenticated, and the processing continues to the next step being authorization.

4.10.2.2. Endpoint Authorization

The Broker authorizes access to its message queues in order to protect the queues from unauthorized access. The following authorization rules are defined:

- An Endpoint can only consume messages from its own message queue, i.e. queue named by the given Endpoint containing messages where that Endpoint is the receiver.
- An Endpoint can upload messages to any message queue.
- An Endpoint can send a message to the Broker only if the Broker permits transfer of message with given message type and for given receiver (see Restricting Message Paths).

4.10.2.3. Authentication or Authorization Failure

If either the authentication or the authorization fails, the Broker refuses the connection, or the operation requested by the Endpoint and returns a security error to the client. The error contains message with a brief reason for not granting the access.

5. Appendix A: Glossary

Acronym / Keyword	Description
BA	Business Application
ECP	Energy Communication Platform. The reference implementation of MADES.
ECP component	A key functional block of the ECP architecture. There are three types: Endpoint, Broker and Component Directory.
FSSF	File System Shared Folder. The file-based channel for accessing the ECP public API.
MADES	Market Data Exchange Standard.
UC / Use Case	A synonym for “functionality” – either visual or nonvisual.

6. Appendix B: ECP Logo

Part of the ECP delivery is a set of ECP logo images that are available in an attachment [ECP Logo]. Below is the logo in default colour scheme.

Figure 62: ECP Logo

The attachment contains images in two colour models, each meant for different usage:

- RGB model – meant for usage in electronic documents such as web pages
- CMYK model – suitable for printed material Images are delivered in two formats:
  - AI format – vector graphics format allowing to perform transformations such as scaling without the loose of quality
  - PNG format – raster graphics format used for general purposes

In addition, the images are delivered in three-colour forms. Which one to use depends on the colour context in which the logo is used:

- Image in default colours as shown above on the thumbnail
- Image with inverted colours
- Image in grayscale

7. Attachments

Attachment	Filename	Description
Internal API	06_ECP4 Internal API schemas.zip	XSDs for Internal API.
ECP Logo	07_ECP4 Logo.zip	Images with ECP logo.

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